

Unit 8

DATABASES

30 items · 401 marks

Syllabus sections in this unit

8.1 Database Concepts

8.2 Database Management Systems (DBMS)

8.3 Data Definition Language (DDL) and Data Manipulation Language (DML)

Unit 8: Databases

1. M/J 26/P11/Q4/a,b,c

- 4 A shop allows customers to pay to borrow toys. The shop uses a database to record information about the customers and the toys they have borrowed.

CUSTOMER (CustomerID, ContactFirstName, Phone)

TOY (ToyID, ToyName, CostPerDay)

TOY_INSTANCE (UniqueToyID, ToyID, Quality)

BORROW (BorrowID, CustomerID, UniqueToyID, StartDate, NumberDays)

- (a) Complete the table by describing the relationship between each pair of tables. Include the primary key(s) and foreign key(s) and how these create each relationship.

pair of tables	description of relationship
CUSTOMER and BORROW	
TOY_INSTANCE and TOY	

[4]

- (b) Write a Structured Query Language (SQL) script to return the total number of days of all toys borrowed by each customer. The customer ID and first name for each customer also need to be returned.

The result needs to include a suitable field name for the total number of days.

[4]

- (c) The table BORROW needs amending to include an additional field.

- (i) The field will store whether the toy has been returned and will store either Yes or No.

Write an SQL script to include the additional field in the table BORROW.

[2]

- (ii) The toy that was borrowed with the borrow ID A2226 has been returned.

Write an SQL script to amend the record and store that the toy has been returned.

[3]

2. M/J 26/P12/Q3/a,b,c

- 3** A computer game developer stores data about a computer game in a database. The database has three entities:

- the users, who are called players
- the characters that users can create
- the groups that users can create and allow characters to join.

(a) Describe the relationships between the **three** entities.

[2]

(b) The database is created with **four** tables:

PLAYER(PlayerID, PlayerName, DateJoined)

CHARACTER(CharacterID, CharacterName, DateCreated, Level, NumberOfPoints, PlayerID)

GROUP(GroupID, GroupName, Category)

CHARACTER_GROUP(CharacterGroupID, CharacterID, GroupID, DateJoinedGroup)

Sample data for the table CHARACTER is given:

CharacterID	CharacterName	DateCreated	Level	NumberOfPoints	PlayerID
00123	Crystal	05/05/2025	5	35687	85AD
00124	Diamond	06/06/2025	4	21055	69CD
00125	Ruby	07/10/2025	1	1002	85AD

- (i) The given Structured Query Language (SQL) script is written to define the table CHARACTER.

```

01  CREATE TABLE CHARACTER(
02      CharacterID INT,
03      CharacterName VARCHAR,
04      DateCreated DATE,
05      TheLevel INT,
06      NumberOfPoints INT,
07      PlayerID VARCHAR,
08      PRIMARY KEY CharacterID,
09      FOREIGN KEY (PlayerID)
10  );

```

The SQL script contains **four** errors.

Give the line number of each error and the corrected line.

error line 1

corrected line

error line 2

corrected line

error line 3

corrected line

error line 4

corrected line

[4]

- (ii) Write an SQL script to return each player's name and the number of characters each player has created.

The result needs to include a suitable field name for the number of characters.

Order the results in descending order of player ID.

[5]

- (c) Complete the description of database key terms by writing the missing words.

The stores data about the data in the database. The table names, field names and relationships will be stored. Two other examples of data that will be stored are and

When designing a database, the schema is the design of the database tables, fields and relationships independent of any Database Management System (DBMS).

A DBMS provides security features to protect the data.

are used to give user accounts permission to read data, some to read and write data and others may be unable to read the data.

When creating a database, a provides tools to create visual components for example forms and reports.

[5]

3. M/J 26/P13/Q4/a,b,c

- 4 A database needs designing for a school library.

The library stores data about books including the author, title, category and number of pages.

The library will also store data about the students and the books they have borrowed. The data needs to include the student's name and student's class.

Books are borrowed for 14 days. A book is late if it has not been returned within the 14 days. The database needs to store whether the book is late, or not late.

- (a) Create a 3-table design for the library database normalised to Third Normal Form (3NF).

Give your table design in the format:

TableName (Field1, Field2, Field3, ...)

A primary key is underlined with a solid line.

A foreign key is underlined with a broken line.

table 1

table 2

table 3

[6]

(b) Explain the reasons why the library database should enforce referential integrity.

[2]

(c) The school is also creating a database to store data about the departments, the staff and classes in the school. The database is called `SCHOOLDATA`. The database has four tables:

`STAFF(StaffID, FirstName, FamilyName, DepartmentID)`

`DEPARTMENT(DepartmentID, DepartmentName, DepartmentLeader)`

`CLASS(ClassID, YearGroup, SubjectCode)`

`SUBJECT(SubjectCode, SubjectName, DepartmentID)`

(i) The table shows some sample data for `SUBJECT`.

SubjectCode	SubjectName	DepartmentID
CS1	Computer Science IGCSE	ITCS
CS2	Computer Science IAS-level	ITCS
MA3	Mathematics IGCSE	MTS
ME4	Media IGCSE	ITCS

The Structured Query Language (SQL) scripts to create the database and the table `SUBJECT` are shown.

```

01  CREATE DATA BASE SCHOOLDATA;
02  CREATE TABLE SUBJECT(
03      SubjectCode VARCHAR NONULL,
04      SubjectName VARCHAR
05      DepartmentID VARCHAR,
06      PRIMARY KEY (SubjectCode),
07      FOREIGN KEY (DepartmentID) REFERENCES (DepartmentID)
08  );

```

The SQL scripts contain **four** errors.

Give the line number of each error and the corrected line.

error line 1

corrected line

error line 2

corrected line

error line 3

corrected line

error line 4

corrected line

[4]

- (ii) Write an SQL script to return each department's name and the number of subject codes for that department.

The result needs to include a suitable field name for the number of subject codes.

Order the results in ascending order of department name.

[5]

4. O/N 25/P11/Q2/a,b,c,d

- 2 A relational database, `WORKEXP`, stores data about students and the companies where they complete work experience.

Students can complete multiple work experience placements but can only complete one placement at a time.

Students can complete more than one placement at the same company.

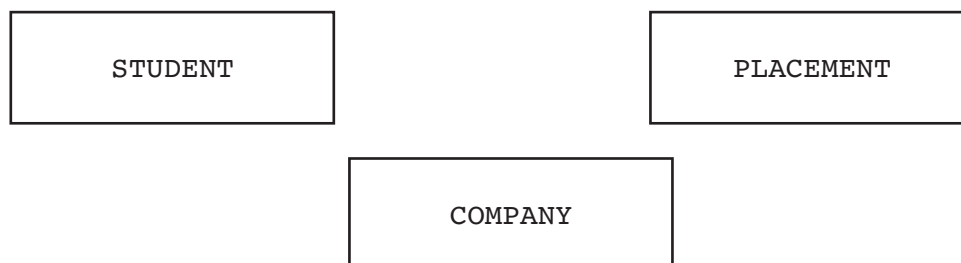
Part of the database is shown:

`STUDENT`(`StudentID`, `FirstName`, `TelephoneNumber`, `UniversityName`)

`PLACEMENT`(`PlacementID`, `StudentID`, `CompanyID`, `StartDate`, `EndDate`, `Complete`)

`COMPANY`(`CompanyID`, `CompanyName`, `MaxStudentsPerPlacement`)

- (a) Complete the entity-relationship (E-R) diagram for the database `WORKEXP`.



[2]

- (b) Explain why the database `WORKEXP` is in Third Normal Form (3NF).

[2]

- (c) Some example data from the `PLACEMENT` table is shown:

<code>PlacementID</code>	<code>StudentID</code>	<code>CompanyID</code>	<code>StartDate</code>	<code>EndDate</code>	<code>Complete</code>
PC001	LDEA01	MSCM	01/01/2023	12/07/2023	TRUE
PC002	LDEA01	MSCM	01/04/2024	08/05/2024	FALSE
PC003	ALAU02	NEAM	09/03/2020	10/03/2021	FALSE
PC004	LOLI75	GOUZ	07/06/2018	11/09/2018	TRUE

- (i) Write a Structured Query Language (SQL) script to delete all placements that have been completed.

[2]

- (ii) Write an SQL script to return the total number of placements completed by the student with ID LDEA01 at the company with ID NEAM. The total should be given an appropriate name.

[4]

- (d) The database tables are repeated here:

STUDENT(StudentID, FirstName, TelephoneNumber, UniversityName)

PLACEMENT(PlacementID, StudentID, CompanyID, StartDate, EndDate,
Complete)

COMPANY(CompanyID, CompanyName, MaxStudentsPerPlacement)

Explain what is meant by referential integrity, and how it applies to this database.

[3]

5. O/N 25/P12/Q4/a,b,c,d

- 4 A relational database, SHIPPING, stores data about the ships in a company and the containers that are carried on the ships.

The database has the following tables:

CONTAINER(ContainerID, Type, Weight, OwnerName, ShipID)

SHIP(ShipID, Type, Capacity, ShipName)

- (a) Describe the relationship between the two tables. Refer to the primary and foreign keys in your answer.

[2]

- (b) The table CONTAINER needs an additional field to store the data for the last inspection date.

Write a Structured Query Language (SQL) script to add **one** field to the table CONTAINER to store the date of last inspection of the container, for example 08/07/2019.

[2]

- (c) Write an SQL script to return the number of containers stored in the database for the ship with the name Caledonia.

[4]

- (d) Describe the purpose of a developer interface in a Database Management System (DBMS).

[2]

6. O/N 25/P13/Q5/a,b,c,d

5 A company creates a relational database to store data about its customers.

The database, `REVIEWS`, stores data about the customers, products, complaints and staff.

The database has four tables:

`CUSTOMER`(`CustomerID`, `CustomerFirstName`, `CustomerLastName`, `CustomerEmail`)

`PRODUCT`(`ProductID`, `ProductName`, `ProductDetail`, `Price`, `Rating`)

`COMPLAINT`(`ComplaintID`, `ProductID`, `CustomerID`, `ComplaintDetails`, `StaffID`)

`STAFF`(`StaffID`, `StaffFirstName`, `StaffLastName`, `Department`, `RemoteWorker`)

(a) Complete the entity-relationship (E-R) diagram for the database `REVIEWS`.



[3]

(b) Some example data from the `STAFF` table is shown.

<code>StaffID</code>	<code>StaffFirstName</code>	<code>StaffLastName</code>	<code>Department</code>	<code>RemoteWorker</code>
1	Ralph	Jura	E	Yes
2	Luca	Emcee	A	Yes
3	Darwin	Acula	F	No

Write a Structured Query Language (SQL) script to define the table `STAFF`.

[4]

(c) Products are given a rating between 1 and 10 inclusive.

Write an SQL script to return only the `ProductID`, `ProductName` and `ComplaintDetails` for all products with a rating of 5 or less. The results need to be displayed in descending order of rating.

[5]

(d) A Database Management System (DBMS) is used to maintain and manage the database.

Describe **two** ways in which the DBMS can be used to ensure the security of the customer data.

1

2

[4]

7. M/J 25/P11/Q5/a,b,c,d,e

- 5** A company builds and sells furniture to customers. The company stores data about customers, their payment cards and their furniture orders in a database.

The database, FURNITURE, has the following tables:

CUSTOMER(CustomerID, Name, Phone)

CUSTOMER_CARD_DATA(CardID, CustomerID, CardType, CardNumber, EndDate)

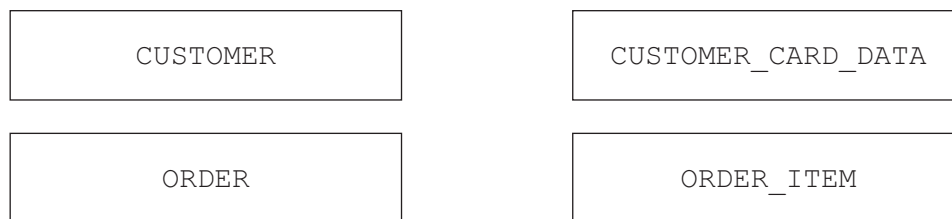
ORDER(OrderID, CustomerID, TotalCost, Paid, OrderDate, Complete)

ORDER_ITEM(OrderItemID, OrderID, Type, Height, Width, Depth, Details)

The primary keys are underlined in each table.

The attribute `Complete` in the table `ORDER` stores the Boolean value `TRUE` if the order has been built and `FALSE` if the order has not been built.

- (a) Complete the entity-relationship (E-R) diagram for the database.



[3]

- (b) Identify **one** attribute in the table `CUSTOMER_CARD_DATA` that could be a candidate key.

[1]

- (c) Identify **two** tables in the database that contain one or more foreign keys.
Give **one** attribute that is a foreign key in each table.

	Table	Foreign key
1		
2		

[2]

- (d) Explain the reasons why the data in the table `ORDER_ITEM` cannot be stored in the table `ORDER`

[3]

- (e) Write an Structured Query Language (SQL) script to output the customer ID, the customer's name and the total cost of the customer's orders that have **not** been paid.

The output of the total cost must have an appropriate title.

[4]

8. M/J 25/P12/Q5/a,b,c,d

- 5 An online game has a database that stores data about users and the characters each user creates in the game. Each user can create multiple characters and purchase multiple items for each character.

The normalised database has the following design:

USER (Username, Password, DateOfBirth)

CHARACTER (CharacterName, CharacterID, Username, Level, Money)

ITEM (ItemName, MinimumLevel, Cost)

CHARACTER_ITEM (CharacterID, ItemName)

- (a) Explain the purpose of the table CHARACTER_ITEM in the database.

[2]

- (b) Underline the attribute, or attributes, that form the primary key in each of the tables.

USER (Username, Password, DateOfBirth)

CHARACTER (CharacterName, CharacterID, Username, Level, Money)

ITEM (ItemName, MinimumLevel, Cost)

CHARACTER_ITEM (CharacterID, ItemName)

[2]

- (c) A Database Management System (DBMS) provides data security.

- (i) Identify **two** methods the DBMS can use to protect the data in the table USER from unauthorised access.

Explain how each method protects the data.

Method 1

Explanation

Method 2

Explanation

[4]

- (ii) The DBMS also supports data integrity.

Give **two** ways that a DBMS can support data integrity.

1

2

[2]

- (d) (i) Write a Structured Query Language (SQL) script to count the number of items purchased by the user with the username "KAT123".

[3]

- (ii) The following changes need to be made to the character with the ID "0002":

- level changed to 3
- money changed to 10000.00

Write an SQL script to change the character's data.

[3]

9. M/J 25/P13/Q6/a,b,c,d

- 6** A shop sells pens to customers. Customers place an order with the shop and collect the items the next day. The shop uses a database to store the information about the orders.

The database contains the following tables:

CUSTOMER(CustomerID, CustomerName, Email)

ORDER(OrderID, CustomerID, Date, Collected)

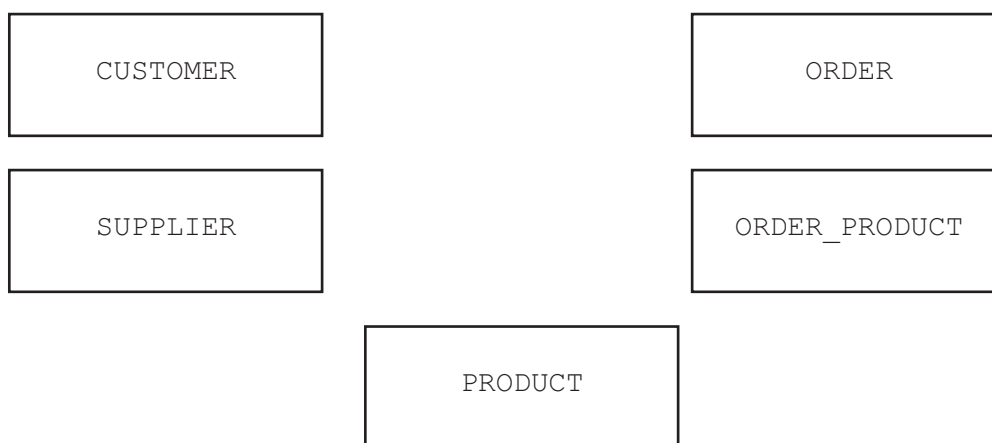
ORDER_PRODUCT(OrderID, ProductID, Quantity)

PRODUCT(ProductID, ProductName, QuantityInBox, Cost, SupplierID)

SUPPLIER(SupplierID, SupplierName, SupplierEmail)

The primary keys are underlined.

- (a)** Complete the entity-relationship (E-R) diagram for the database.



[4]

- (b)** A new product needs to be entered into the database. The product has the ID 002323, the product name 'Blue ball point 2mm', there are 50 in a box, the product costs \$5.00 and the supplier has the ID SFX223.

Write a Structured Query Language (SQL) script to enter the new product into the table PRODUCT.

[4]

- (c)** The attribute `Collected` in the table `ORDER` stores the Boolean value `TRUE` if the order has been collected and `FALSE` if the order has not been collected.

Write an SQL script to return the customer name for each customer that has orders they have not collected. Include the number of orders each customer has not collected with an appropriate title.

An example output might be:

CustomerName	NotCollected
Jack Wright	2
Lin Cho	1
Santaya Yui	1

[4]

(d) The shop uses a Database Management System (DBMS).

(i) One feature provided by a DBMS is a data dictionary.

The data dictionary stores the attribute names, table names, foreign keys and primary keys in the database.

Identify **three** other items stored in a data dictionary.

1

2

3

[3]

(ii) The DBMS provides a developer interface.

Explain how a database designer can make use of the developer interface.

[3]

10. O/N 24/P11/Q2/a,b,c

2 A shop repairs electronic devices, for example mobile phones and tablet computers. The shop owner stores the data about the repairs using a file-based approach.

(a) Give **one** limitation of using a file-based approach to store the data **and** explain how a relational database addresses this limitation.

Limitation

Explanation

[3]

(b) The shop owner creates a relational database called `FIXIT`.

The database stores data about the customers and the devices for repair.

Some devices need new parts that are ordered from suppliers.

The database `FIXIT` is designed to include the following tables:

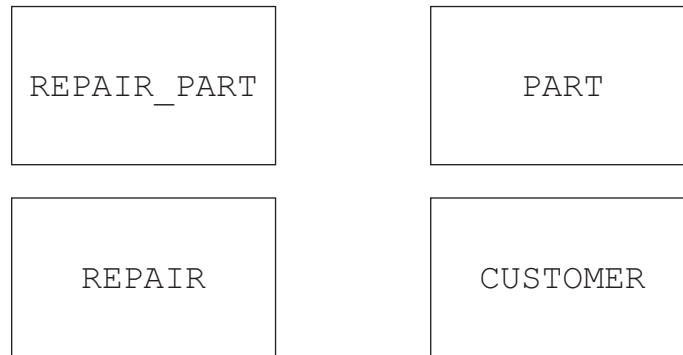
`PART(PartID, Description, Price, SupplierID)`

`CUSTOMER(CustomerID, FirstName, LastName, ContactNumber)`

`REPAIR(RepairNumber, StartDate, EndDate, CustomerID, Device)`

`REPAIR_PART(PartID, RepairNumber, Quantity)`

- (i) Complete the entity-relationship (E-R) diagram for the given tables.



[3]

- (ii) The table shows sample data for the table `REPAIR_PART`.

PartID	RepairNumber	Quantity
ACD128SA	0022	3
PPOR543DWW	0022	1
TR453	0023	1
PPOR543DWW	0023	2
WED5	0024	5

Write a Structured Query Language (SQL) script to define the table `REPAIR_PART`.

Include constraints (restrictions) on the data that can be entered into each field where appropriate.

[5]

- (iii) Suppliers send invoices to the company for the parts that are used. A new table, `INVOICE`, stores the data about each invoice and whether it has been paid or not.

The design for the table `INVOICE` is shown:

```
INVOICE(InvoiceID, SupplierID, AmountDue, Paid, DatePaid)
```

The table shows sample data for the table `INVOICE`.

InvoiceID	SupplierID	AmountDue	Paid	DatePaid
000001	JK675	22.50	TRUE	01/01/2024
000002	WR443	358.99	FALSE	
000003	JK675	10.21	FALSE	

Write an SQL script to return the total amount due to the supplier with the ID of JK675 for all the invoices that have **not** currently been paid.

[3]

(c) Complete the table by writing a definition for each of the database terms.

Term	Definition
Referential integrity	
Candidate key	
Tuple	

[3]

11. O/N 24/P12/Q6/a,b

- 6** A company uses a relational database to store data about its customers, employees and the individual repair jobs that customers have booked.

(a) Explain the benefits of using a relational database instead of a file-based approach.

[3]

- (b)** The company decides which employees will work on each repair job. An employee can log into the database to access information about their repair jobs.

The database is normalised and includes these tables:

- CUSTOMER stores personal data about each customer
- EMPLOYEE stores personal data about each employee
- LOGIN_DATA stores the username and password for each employee
- JOB stores the data about each repair job
- JOB_EMPLOYEE stores the employees that are working on each repair job.

- (i)** Identify each relationship between the database tables **and** explain how each relationship can be implemented in the normalised database.

[6]

- (ii)** The database also has the table INVOICE that stores data about each invoice that is sent to a customer.

Example data from the table INVOICE is given.

InvoiceID	DateSent	Amount	Paid	JobID
29262	12/12/2023	105.20	Y	221
26765	11/11/2023	200.00	Y	315
13290	02/01/2024	50.00	Y	315
34090	05/02/2024	25.95	N	569

Write a Structured Query Language (SQL) script to return the total amount of all the invoices sent in the year 2023 that have been paid.

[3]

12. O/N 24/P13/Q4/a,b,c,d,e

- 4** A company makes ice cream and sells it to shops.

The ice cream is made in batches: a large quantity of one type and flavour of ice cream that is then split into smaller quantities for sale.

The company's owner has designed a relational database, ICECREAM, to store data about their ice cream and customers.

Some of the tables in the database are given. The database is not normalised.

BATCH(BatchID, Type, Flavour, Size, SellingPrice, EndDate)

CUSTOMER(CustomerID, CompanyName, EmailAddress, TelephoneNumber)

SALE(SaleID, BatchID, CustomerID, Quantity, Date)

- (a)** Identify **two** foreign keys in the table SALE **and** the table that each foreign key references.

Foreign key 1

Table name 1

Foreign key 2

Table name 2

[2]

- (b) Write an SQL script to return the total quantity of ice cream sold to the customer with the ID of 0034E in the year 2023.

[3]

- (c) The table definition for `BATCH` is repeated here:

`BATCH(BatchID, Type, Flavour, Size, SellingPrice, EndDate)`

Sample data for the table `BATCH` is given:

BatchID	Type	Flavour	Size	SellingPrice	EndDate
KIV12	Plain	Vanilla	1	2.20	12/12/2024
RIC14	Plain	Chocolate	0.5	1.80	12/12/2024
TYL1	Non-dairy	Lemon	0.5	2.10	01/02/2025
FYV2	Non-dairy	Vanilla	0.25	1.50	02/02/2025
BIV13	Plain	Vanilla	1	2.20	02/02/2024

- (i) Write an SQL script to define the table `BATCH`.

Include constraints (restrictions) on the data that can be entered into each field where appropriate.

[5]

- (ii) The table `BATCH` is not normalised.

Normalise the database table `BATCH`.

Write the table definitions for your new tables.

Identify any primary and foreign keys in your tables.

Do **not** change or include the tables `CUSTOMER` and `SALE`.

[4]

(d) Complete the following table by defining each database term.

Database term	Definition
Entity	
Attribute	

[2]

(e) A Database Management System (DBMS) supports data integrity.

Explain how a DBMS supports data integrity.

[3]

13. M/J 24/P11/Q6/a,b,c

- 6 A company is developing a website that will allow users to create an account and then play a quiz every day. The data about the users and the quizzes are stored in a database.

A user must select a unique username and enter a valid email address to create an account. All users must be over the age of 16. A new quiz is given to the users every day. Each quiz is stored in its own text file.

The database stores the filename of each quiz and the date it can be played. The user gets a score for each quiz they complete, which is stored in the database. The scores are used to give each user a rating, for example Gold.

(a) Create a 3-table design for this database normalised to Third Normal Form (3NF).

Give your table design in the format:

TableName (PrimaryKey, Field1, Field2, ...)

[6]

(b) The company is using a Database Management System (DBMS) to set up the database.

Describe what is meant by the following DBMS features:

Data dictionary

Logical schema

[4]

(c) The company has another database, FARMING, for a different game.

The database FARMING has a table named EVENT which is shown with some sample data.

PlayerID	EventID	Category	Points
000123	3	Build	100
000124	1	Grow	36
000123	4	Grow	22
000123	7	Create	158
000125	3	Grow	85
000125	4	Build	69

- (i) The database `FARMING` has a second table created named `PLAYER` that has the primary key `PlayerID`.

The field `PlayerID` in `EVENT` needs to be set up as a foreign key to link to `PlayerID` in `PLAYER`.

Write a Structured Query Language (SQL) script to change the table definition for `EVENT` to link the foreign key to `PLAYER`.

[2]

- (ii) Write an SQL script to return the number of events that each player has completed.

[3]

14. M/J 24/P12/Q4/a,b,c,d

- 4** An assessment board wants to store the marks students achieved in exams in a database named RECORDS.

Part of the database design includes these two tables:

EXAM(ExamID, Subject, Level, TotalMarks)

EXAM_QUESTION(ExamQuestionID, ExamID, QuestionNumber, Question, MaxMark)

- (a)** Identify the relationship between EXAM and EXAM_QUESTION.

[1]

- (b)** Sample data for the table EXAM is shown:

ExamID	Subject	Level	TotalMarks
00956124	Computer Science	2	75
00956125	Computer Science	3	120
00956126	Mathematics	2	100
00956127	Mathematics	3	150
00956128	Physics	2	70
00956129	Physics	3	80

Write a Structured Query Language (SQL) script to define the table EXAM.

[3]

- (c)** The table EXAM_QUESTION has been created but the foreign key has not been linked.

Write an SQL script to update EXAM_QUESTION **and** link the foreign key to EXAM.

[2]

- (d)** The database also needs to store data about the students, the exams the students have taken and the marks the students achieved in each question of each exam.

Describe the additional tables that will need to be included in the database **and** explain how all the tables in the database will be linked.

[5]

15. M/J 24/P13/Q4/a,b,c,d

- 4** A theatre wants to use a database to store data about the shows that are scheduled, their customers and the seats that the customers have booked.

In the theatre:

- Each show can take place on multiple dates.
- Some dates can have more than one performance.
- There are multiple rows of seats.
- Each seat can be individually booked by its row letter and seat number, for example row E seat 2.

Part of the database design includes these tables:

SHOW(ShowID, Title, Duration)

SEAT(SeatID, RowLetter, SeatNumber)

PERFORMANCE(PerformanceID, ShowID, ShowDate, StartTime)

- (a)** Identify the relationship between the tables PERFORMANCE and SHOW.

[1]

- (b)** Sample data for the table PERFORMANCE is shown:

PerformanceID	ShowID	ShowDate	StartTime
0001	MK12	5/5/2025	13:00
0002	MK12	5/5/2025	19:30
0003	MK12	6/5/2025	19:00
0004	OP3	7/5/2025	18:30
0005	OP3	8/5/2025	18:30
0006	OP3	9/5/2025	13:00

Write a Structured Query Language (SQL) script to define the table PERFORMANCE.

[4]



- (c)** Write an SQL script to return the number of times each show is scheduled. For example, in the sample data in part **(b)**, the show MK12 is scheduled three times.

The result needs to include the show name and a suitable field name for the number of times it is scheduled.

[4]

- (d)** Customers give their first name, last name and email address when they are making a booking. One booking can include multiple seats.

Describe the additional tables that will need to be included in the database **and** explain how these tables will be linked within the database.

[5]



16. O/N 23/P11/Q3/a,b,c

- 3** A shop manager has designed a relational database to store customer orders.

The database will have the following tables:

CUSTOMER(CustomerID, FirstName, LastName, Town)

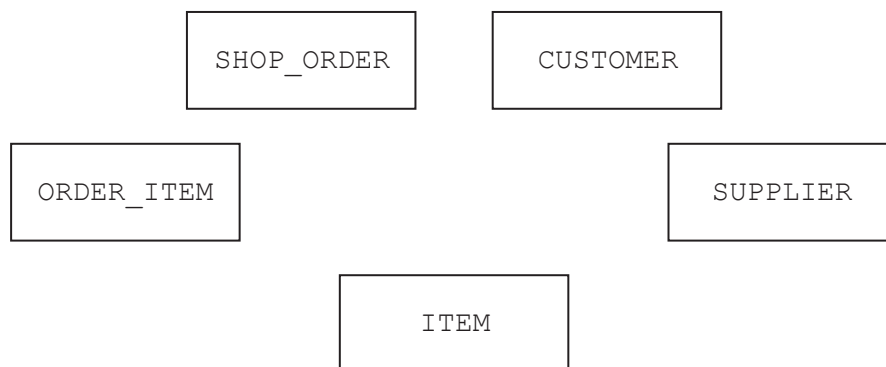
SHOP_ORDER(OrderNo, CustomerID, OrderDate)

SUPPLIER(SupplierID, EmailAddress, TelephoneNumber)

ITEM(ItemNumber, SupplierID, Description, Price)

ORDER_ITEM(ItemNumber, OrderNo, Quantity)

- (a)** Complete the entity-relationship (E-R) diagram for the relational database.



[3]

- (b)** Identify **three** advantages of a relational database compared to a file-based approach.

1

2

3

[3]

- (c) (i)** Write a Structured Query Language (SQL) script to define the database called *SHOP*.

[1]

- (ii)** Write the SQL script to return the total quantity of items that the customer with the ID of HJ231 has ordered.

[4]

17. O/N 23/P12/Q2/a,b,c

- 2 (a)** State what is meant by the following terms in a relational database model.

Entity

Primary key

Referential integrity

[3]

- (b)** Authentication is one method a Database Management System (DBMS) can use to improve the security of a database.

Describe **other** methods that a DBMS can use to improve the security of a database.

[4]

- (c)** The following database table is not normalised.

StudentName	DateOfBirth	TutorGroup	Subject	SubjectCode
Yuwei Chen	01/09/2004	SMH	English, Maths, Computer Science	EN, MA, CS
Claudia Raj	23/02/2005	JMB	Maths, Physics, Art	MA, PY, AR
Aamil Akram	24/01/2005	KMB	Art, Design, English language	AR, DE, EN
Areeba Faraz	21/12/2004	SMH	English language, Chemistry, Design	EN, CH, DE

Explain how to modify the table to put it into First Normal Form (1NF).

[4]

18. O/N 23/P13/Q3/b

- 3** A company sells online Computer Science courses to students in different countries.

The courses are stored on a public cloud.

- (b)** The company uses a database, COURSES, to store data about the courses and their tutors.

Each course starts at different times of the year and may have a different tutor.

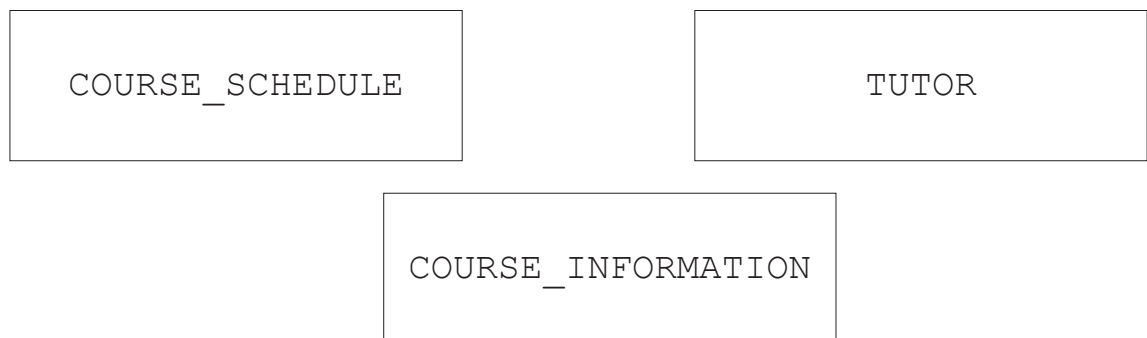
The database has the following structure:

COURSE_INFORMATION(CourseID, Description, Cost)

TUTOR(TutorID, TelephoneNumber, EmailAddress, TutorName)

COURSE_SCHEDULE(CourseID, DateStarted, TutorID)

- (i)** Complete the entity-relationship (E-R) diagram for the database COURSES.



[1]

- (ii)** Write the Structured Query Language (SQL) script to return the total number of courses that have started after 9 September 2023.

The value returned must have an appropriate field name.

[4]

19. M/J 23/P11/Q2/a,b

- 2** An organisation uses a database to store data about the types of bird that people have seen.

- (a)** The database is managed using a Database Management System (DBMS).

- (i)** State what is meant by a data dictionary **and** give **one** example of an item typically found in a data dictionary.

Definition

Example

[2]

- (ii)** State what is meant by data integrity **and** give **one** example of how this is implemented in a database.

Definition

Example

[2]

- (b) The database, `Birds`, stores information about the types of bird and the people who have seen them.

Data about each bird seen is stored with its location and data about the person who saw the bird.

Database `Birds` has the following tables:

```
BIRD_TYPE(BirdID, Name, Size)
```

```
BIRD_SEEN(SeenID, BirdID, Date, Location, PersonID)
```

```
PERSON(PersonID, FirstName, LastName, EmailAddress)
```

- (i) Complete the table by identifying **two** foreign keys and the database table where each is found.

Foreign key	Database table

[2]

- (ii) The database `Birds` has been normalised.

Draw **one** line from each Normal Form to the most appropriate definition.

Normal Form

Definition

First Normal Form (1NF)

All fields are fully dependent on the primary key.

Second Normal Form (2NF)

There are no repeating groups of attributes.

Third Normal Form (3NF)

There are no partial dependencies.

[1]

- (iii) Part of the database table `BIRD_TYPE` is shown:

BirdID	Name	Size
0123	Blackbird	Medium
0035	Jay	Large
0004	Raven	Large
0085	Robin	Small

The database only supports these data types:

- character
- varchar
- Boolean
- integer
- real
- date
- time

Write a Structured Query Language (SQL) script to define the table `Bird_Type`.

[4]

(iv) The database tables are repeated here for reference:

```
BIRD_TYPE(BirdID, Name, Size)
```

```
BIRD_SEEN(SeenID, BirdID, Date, Location, PersonID)
```

```
PERSON(PersonID, FirstName, LastName, EmailAddress)
```

Complete the SQL script to return the number of birds of each size seen by the person with the ID of J_123.

```
SELECT BIRD_TYPE.Size, ..... (BIRD_TYPE.BirdID)

      AS NumberOfBirds

FROM BIRD_TYPE, .....

WHERE ..... = "J_123"

AND BIRD_TYPE.BirdID = .....

..... BIRD_TYPE.Size;
```

[5]

20. M/J 23/P12/Q2/a,b,c(ii,iii)

2 A horse riding school uses a database, `Lessons`, to store data about lesson bookings.

This database is created and managed using a Database Management System (DBMS).

(a) The table contains names and descriptions of DBMS features and tools.

Complete the table by writing down the missing names and descriptions.

Name	Description
Data dictionary	
Query processor	
.....	A model of a database that is not specific to one DBMS.
.....	A software tool that allows the user to create items such as tables, forms and reports.

[4]

(b) Explain the reasons why referential integrity is important in a database.

[3]

(c) The database `Lessons` has the following tables:

```
HORSE(HorseID, Name, Height, Age, HorseLevel)
```

```
STUDENT(StudentID, FirstName, LastName, RiderLevel, PreferredHorseID)
```

```
LESSON(LessonID, Date, Time, StudentID, HorseID, LessonContent)
```

Dates in this database are stored in the format `#DD/MM/YYYY#`.

The fields `RiderLevel` and `HorseLevel` can only have the values: `Beginner`, `Intermediate` or `Advanced`.

(ii) Write a Structured Query Language (SQL) script to return the names of all the horses that have the horse level intermediate or beginner.

[4]

(iii) The following SQL script should return the number of riders that have the rider level beginner and have a lesson booked on 09/09/2023.

```
SELECT SUM(STUDENT.RiderLevel) AS NumberOfRiders  
  
FROM STUDENT, LESSON  
  
WHERE StudentID = StudentID  
  
OR Date = #09/09/2023#  
  
AND STUDENT.RiderLevel = Beginner;
```

There are **four** errors in the script.

Identify **and** correct each error.

1

2

3

4

[4]

21. M/J 23/P13/Q4/a,b,c

- 4** A shop rents cars to customers. The shop uses a relational database to store information about the rentals.

(a) Describe **two** ways in which a relational database addresses the limitations of a file-based approach.

1

2

[4]

(b) Complete the table by writing the missing term or description for each database feature.

Term	Description
.....	An object that data is stored about.
Tuple	
Secondary key	
.....	A field in one table that is linked to a primary key in another table.

[4]

(c) The car rental database is not normalised. The current database design is:

```
BOOKING (CarRegistration, StartDate, EndDate,
         CarModel, CarColour, CustomerFirstName)
```

```
CUSTOMER (CustomerFirstName, CustomerLastName, EmailAddress,
          TelephoneNumber)
```

Write a normalised database design for this database.

All tables must be in Third Normal Form (3NF).

Use the field names given **and** underline the primary key fields.

[4]

22. O/N 22/P11/Q4/a,b,c,d

- 4 A photographer creates a relational database to store data about photographs taken at birthday parties.

The database, PHOTOGRAPHS, stores details of the customer, the party, the photographs taken and the cameras used.

The photographer has several cameras that are used for taking the photographs at the parties.

Each camera has a specific lens type (for example, XY32Z) and lighting type (for example, F1672).

Data about each photograph is stored in the database including the party at which it was taken, the time it was taken and the camera used.

The database has these four tables:

CUSTOMER(CustomerID, FirstName, LastName, Telephone)

PARTY(PartyID, CustomerID, PartyDate, StartTime)

PHOTO_DATA(PhotoID, PartyID, TimeTaken, CameraID)

CAMERA_DATA(CameraID, LensType, LightingType)

(a) Complete the entity-relationship (E-R) diagram for the database PHOTOGRAPHS.



[3]

(b) The database is normalised and is in Third Normal Form (3NF).

Describe the characteristics of a database that is in Third Normal Form (3NF).

[3]

(c) The table shows some sample data for the table PHOTO_DATA.

PhotoID	PartyID	TimeTaken	CameraID
ST23-56	BD987	08:34	NIK-02
ST23-57	BD987	08:55	NIK-02
ST23-60	BC08	09:01	CAN-01
ST23-61	BC08	10:23	CAN-12
ST23-62	BC08	10:56	NIK-01

(i) State what is meant by a **tuple**. Give an example of a tuple from PHOTO_DATA.

Tuple

Example

[2]

(ii) Complete the Structured Query Language (SQL) script to display the total number of photographs that have been taken using a camera with a camera ID starting with CAN.

```

SELECT ..... (.....)

FROM

WHERE CameraID LIKE ;

```

[4]

- (d) Write an SQL script to include two new fields in CAMERA_DATA to store the number of photographs currently on the camera **and** the date the camera was last used.

[3]

23. O/N 22/P12/Q5/a,b,c,d

- 5 A relational database, GARDEN, has the following tables:

OWNER(OwnerID, FirstName, TelephoneNo, TreeID, TreePosition)

TREE(TreeID, ScientificName, MaxHeight, FastGrowing)

- (a) The database is **not** in Third Normal Form (3NF).

Explain how the database can be normalised to 3NF.

[3]

- (b) Write the Structured Query Language (SQL) script to add a new record in the table TREE to store the following data:

Attribute	Value
TreeID	LOW_1276
ScientificName	Salix_Alba
MaxHeight	30.00
FastGrowing	TRUE

[3]

- (c) State what is meant by a **candidate key** in a relational database.

[1]

- (d) (i) Describe, using an example, what is meant by a **data dictionary**.

[2]

- (ii) Describe what is meant by a **logical schema**.

[2]

24. O/N 22/P13/Q2/a,b,c,d,f

- 2** The relational database `ASTRONOMY` is used to store data about telescopes, the companies that own the telescopes and the photographs taken by the telescopes.

The database has these three tables:

`COMPANY (TelephoneNumber, CompanyID, CompanyName)`

`PHOTOGRAPH (PhotoID, TelescopeID, DateTaken, TimeTaken, Elevation)`

`TELESCOPE (TelescopeID, CompanyID, SerialNumber)`

- (a)** Complete the following table by writing the correct answer for each item.

Item	Answer
a suitable field for the primary key in <code>COMPANY</code>	
a candidate key in <code>TELESCOPE</code>	
the degree of relationship between <code>TELESCOPE</code> and <code>PHOTOGRAPH</code>	

[3]

- (b)** A Database Management System (DBMS) has several features.

Identify the feature that describes the relationship between data and its structure.

[1]

- (c)** Complete the SQL script to return the total number of telescopes owned by the company whose ID begins with `HW`.

```
SELECT ..... (.....)
FROM TELESCOPE
WHERE ..... LIKE .....
```

[4]

- (d)** Write the SQL script to add **one** field to the table `PHOTOGRAPH` to store the resolution of the photograph, e.g. `1920 × 1068`.

[2]

- (f)** Describe the purpose of a query processor in a DBMS.

[2]

25. O/N 21/P11/Q5/a,b,c

- 5 Javier owns many shops that sell cars. He employs several managers who are each in charge of one or more shops. He uses the relational database `CARS` to store the data about his business.

Part of the database is shown:

`SHOP(ShopID, ManagerID, Address, Town, TelephoneNumber)`

`MANAGER(ManagerID, FirstName, LastName, DateOfBirth, Wage)`

`CAR(RegistrationNumber, Make, Model, NumberOfMiles, ShopID)`

- (a) Tick (✓) **one** box in each row to identify whether each field is a primary key or a foreign key.

Table	Field name	Primary key	Foreign key
MANAGER	ManagerID		
SHOP	ManagerID		
CAR	RegistrationNumber		
CAR	ShopID		

[2]

- (b) Describe the ways in which access rights can be used to protect the data in Javier's database from unauthorised access.

[3]

- (c) Javier uses Data Definition Language (DDL) and Data Manipulation Language (DML) statements in his database.

- (i) Complete the following DML statements to return the number of cars for sale in each shop.

`SELECT COUNT (.....)`

`FROM .....`

`..... ShopID`

[3]

- (ii) Complete the DML statement to include the following car in the table `CAR`.

Field	Data
RegistrationNumber	123AA
Make	Tiger
Model	Lioness
NumberOfMiles	10500
ShopID	12BSTREET

CAR

..... ("123AA", "Tiger", "Lioness", 10500, "12BSTREET")

[2]

26. O/N 21/P12/Q6/a,b,c

- 6** A shop sells plants to customers. The shop manager has a relational database to keep track of the sales.

The database, PLANTSALLES, has the following structure:

PLANT(PlantName, QuantityInStock, Cost)

CUSTOMER(CustomerID, FirstName, LastName, Address, Email)

PURCHASE(PurchaseID, CustomerID)

PURCHASE_ITEM(PurchaseID, PlantName, Quantity)

(a) The database is normalised.

(i) The table lists the following three stages of normalisation:

- The first stage is from a database that is not normalised (0NF) to First Normal Form (1NF).
- The second stage is from 1NF to Second Normal Form (2NF).
- The third stage is from 2NF to Third Normal Form (3NF).

Tick (✓) **one** box in each row to identify the appropriate stage for each task.

Task	Normalisation stage		
	0NF to 1NF	1NF to 2NF	2NF to 3NF
Remove any partial key dependencies			
Remove any repeating groups of attributes			
Remove any non-key dependencies			

[2]

(ii) Draw an entity-relationship (E-R) diagram for the database PLANTSALLES.



[3]

(b) The shop manager uses a Database Management System (DBMS).

Describe the purpose **and** contents of the data dictionary in the DBMS.

[3]

(c) The shop manager uses both Data Definition Language (DDL) and Data Manipulation Language (DML) statements to create and search the database.

(i) Complete the DML statements to return the total number of items purchased with the purchase ID of 3011A.

SELECT SUM (.....)

FROM

WHERE =;

[4]

(ii) Write DDL statements to include a field in the table `PURCHASE` to store the date of the order.

[3]

27. O/N 21/P13/Q5/a,b,c

5 Javier owns many shops that sell cars. He employs several managers who are each in charge of one or more shops. He uses the relational database `CARS` to store the data about his business.

Part of the database is shown:

`SHOP(ShopID, ManagerID, Address, Town, TelephoneNumber)`

`MANAGER(ManagerID, FirstName, LastName, DateOfBirth, Wage)`

`CAR(RegistrationNumber, Make, Model, NumberOfMiles, ShopID)`

(a) Tick (✓) **one** box in each row to identify whether each field is a primary key or a foreign key.

Table	Field name	Primary key	Foreign key
MANAGER	ManagerID		
SHOP	ManagerID		
CAR	RegistrationNumber		
CAR	ShopID		

[2]

(b) Describe the ways in which access rights can be used to protect the data in Javier's database from unauthorised access.

[3]

(c) Javier uses Data Definition Language (DDL) and Data Manipulation Language (DML) statements in his database.

- (i) Complete the following DML statements to return the number of cars for sale in each shop.

SELECT COUNT (.....)

FROM

..... ShopID

[3]

- (ii) Complete the DML statement to include the following car in the table CAR.

Field	Data
RegistrationNumber	123AA
Make	Tiger
Model	Lioness
NumberOfMiles	10500
ShopID	12BSTREET

CAR

..... ("123AA", "Tiger", "Lioness", 10500, "12BSTREET")

[2]

28. M/J 21/P11/Q7/a,b,c

7 Bobby and Kim are discussing databases.

- (a) Bobby tells Kim that a file-based approach is usually better than a relational database.

Explain why Bobby is incorrect.

[3]

- (b) Bobby has a shop that sells products to customers. His database will store data about his customers, their payment details, orders and the products he sells. Customers will have login details to access their accounts. The database will update customers' payment and login details without keeping any historical records.

- (i) Give **one** example of each of the following relationships from Bobby's database.

one-to-one

one-to-many

many-to-many

[3]

- (ii) Tick (✓) **one** box to identify the relationship that cannot be directly implemented in a normalised relational database.

Relationship	Tick (✓)
one-to-one	
one-to-many	
many-to-many	

[1]

- (iii) Bobby wants to name his database `SHOPORDERS`.

Write a Data Definition Language (DDL) statement to define a new database with the name `SHOPORDERS`.

[1]

- (c) A database has a data dictionary.

Give **three** items that are stored in a data dictionary.

1

2

3

[3]

29. M/J 21/P12/Q1/a,b,c

- 1 Raj owns houses that other people rent from him. He has a database that stores details about the people who rent houses, and the houses they rent. The database, `HOUSE_RENTALS`, has the following structure:

`CUSTOMER`(`CustomerID`, `FirstName`, `LastName`, `DateOfBirth`, `Email`)

`HOUSE`(`HouseID`, `HouseNumber`, `Road`, `Town`, `Bedrooms`, `Bathrooms`)

`RENTAL`(`RentalID`, `CustomerID`, `HouseID`, `MonthlyCost`, `DepositPaid`)

- (a) Give the definition of the following database terms, using an example from the database `HOUSE_RENTALS` for each definition.

Term	Definition and example
Field	
Entity	
Foreign key	

[6]

- (b) Tick (✓) **one** box to identify whether the database `HOUSE_RENTALS` is in Third Normal Form (3NF) or not in 3NF.

Justify your choice using one or more examples from the database `HOUSE_RENTALS`.

In 3NF	
Not in 3NF	

Justification

[2]

- (c) Example data from the table `RENTAL` are given:

RentalID	CustomerID	HouseID	MonthlyCost	DepositPaid
1	22	15B5L	1000.00	Yes
2	13	3F	687.00	No
3	1	12AB	550.00	Yes
4	3	37	444.50	Yes

- (i) Complete the following Data Definition Language (DDL) statement to define the table RENTAL.

```
CREATE ..... (
    RentalID INTEGER NOT NULL,
    CustomerID INTEGER NOT NULL,
    HouseID ..... (5) NOT NULL,
    MonthlyCost ..... NOT NULL,
    DepositPaid BOOLEAN NOT NULL,
    ..... (RentalID)
);
```

[4]

- (ii) Write a Data Manipulation Language (DML) script to return the first name and last name of all customers who have **not** paid their deposit.

[4]

30. M/J 21/P13/Q7/a,b,c

7 Bobby and Kim are discussing databases.

- (a)** Bobby tells Kim that a file-based approach is usually better than a relational database.

Explain why Bobby is incorrect.

[3]

- (b)** Bobby has a shop that sells products to customers. His database will store data about his customers, their payment details, orders and the products he sells. Customers will have login details to access their accounts. The database will update customers' payment and login details without keeping any historical records.

- (i)** Give **one** example of each of the following relationships from Bobby's database.

one-to-one

one-to-many

many-to-many

[3]

- (ii)** Tick (✓) **one** box to identify the relationship that cannot be directly implemented in a normalised relational database.

Relationship	Tick (✓)
one-to-one	
one-to-many	
many-to-many	

[1]

- (iii)** Bobby wants to name his database `SHOPORDERS`.

Write a Data Definition Language (DDL) statement to define a new database with the name `SHOPORDERS`.

[1]

- (c)** A database has a data dictionary.

Give **three** items that are stored in a data dictionary.

1

2

3

[3]

Answers Section

1. M/J 26/P11/Q4/a,b,c

4(a)	1 mark for each correct point - CUSTOMER BORROW is a one-to-many relationship // 1:M - TOY_INSTANCE to TOY is a many-to-one relationship // M:1 - Identifying the correct Primary Key and matching Foreign Key in both relationships - Stating how the keys form the relationship	4	
4(b)	1 mark for each correct point - Selecting the contact name, customer ID and totalling the number of days - needs all 3 - Using an appropriate name - Linking or joining the correct tables - Grouping the results Example 1: <pre>SELECT CUSTOMER.ContactFirstName, CUSTOMER.CustomerID, SUM(NumberDays) AS TotalDays FROM BORROW, CUSTOMER WHERE CUSTOMER.CustomerID = BORROW.CustomerID GROUP BY CUSTOMER.CustomerID;</pre> MP1 MP2 MP3 MP4 Example 2: <pre>SELECT CUSTOMER.ContactFirstName, CUSTOMER.CustomerID, SUM(NumberDays) AS TotalDays FROM BORROW, INNER JOIN CUSTOMER ON CUSTOMER.CustomerID = BORROW.CustomerID GROUP BY CUSTOMER.CustomerID;</pre> MP1 MP2 MP3 MP4	4	CustomerID needs the table reference Accept any correct method of joining the tables, but no mix and match between methods
4(c)(i)	1 mark for each correct statement - Altering the correct table - Adding an appropriately named field with an appropriate data type <pre>ALTER TABLE BORROW ADD Returned BOOLEAN;</pre> accept VARCHAR/CHAR(3)/BOOL	2	Award just 1 mark if statements are in the wrong order or if there are any additional statements Accept any sensible field name
4(c)(ii)	1 mark for each correct statement - Updating the correct table - Setting the additional field to the correct data - Where clause data type must match 4(c)(i) Example: <pre>UPDATE BORROW SET Returned = "Yes" WHERE BorrowID = "A2226";</pre>	3	Award a maximum of 2 marks if statements are in the wrong order or if there are any additional statements

2. M/J 26/P12/Q3/a,b,c

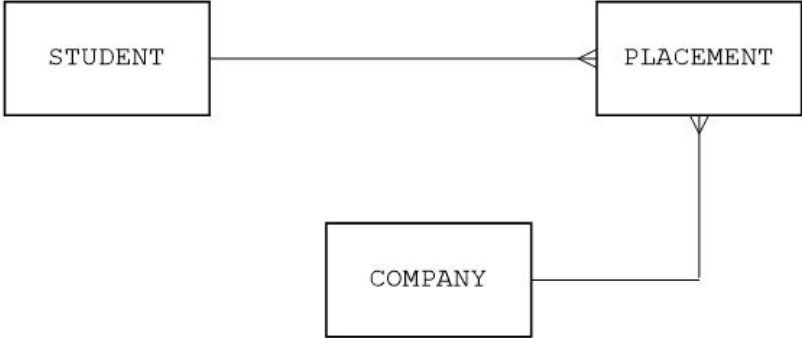
3(a)	1 mark for each correct relationship - The relationship between PLAYER and CHARACTER is one to many /1:M // one player can have many characters, but each character can only be created by one player - The relationship between GROUP and CHARACTER is many to many / M:M // one group can have many characters, and one character can be in many groups	2	The second option in each case needs both 'halves' Accept a m:m relationship between PLAYER and GROUP Accept user for player Mark the writing first, but accept accurate E-R diagram(s), if needed for clarification
------	---	----------	--

3(b)(i)	1 mark for each correct line number and the appropriate correction - Line 02 - CharacterID VARCHAR, // CharacterID VARCHAR (5), - Line 05 Level INT, - Line 08 - PRIMARY KEY (CharacterID) - Line 09 - FOREIGN KEY (PlayerID) REFERENCES PLAYER (PlayerID)	4	Ignore extra NOT NULL on line 02 but not on line 05 or line 08
3(b)(ii)	1 mark for each correct point - Selecting the player's name and counting a field - Using an appropriate name - Linking or joining the correct tables - Grouping the results - Ordering in descending order by player ID Example 1: <pre>SELECT PlayerName, COUNT(CharacterID) MP1 AS TotalCharacters MP2 FROM CHARACTER, PLAYER WHERE CHARACTER.PlayerID = PLAYER.PlayerID MP3 GROUP BY PlayerName MP4 ORDER BY PLAYER.PlayerID DESC; MP5</pre> Example 2: <pre>SELECT PlayerName, COUNT(CharacterID) MP1 AS TotalCharacters MP2 FROM CHARACTER INNER JOIN PLAYER ON CHARACTER.PlayerID = PLAYER.PlayerID MP3 GROUP BY PlayerName MP4 ORDER BY PLAYER.PlayerID DESC; MP5</pre>	5	Accept any correct method of joining the tables, but no mix and match between methods The only attribute that needs a table reference is the PlayerID. Ignore other table references as long as the correct table is used Take care with the DESC
3(c)	1 mark for two data dictionary examples For example: - data types of individual fields/attributes - type of validation - primary key - foreign key 1 mark for each remaining term (4 marks) - data dictionary - logical - access rights - developer interface The data dictionary stores data about the data in the database. The table names, field names and relationships will be stored. Two other examples that will be stored are example 1 and example 2 When designing a database, the logical schema is the design of the database tables, fields and relationships independent of any Database Management System (DBMS). A DBMS provides security features to protect the data. Access rights are used to give user accounts permission to read data, some to read and write data and others may be unable to read the data. When creating a database a developer interface provides tools to create visual components for example forms and reports.	5	Mark the first answer in each space Do not accept table names or field names or relationships, they are given in the question There are many answers for the examples, the list is suggestions only Do not accept metadata as an example or instead of data dictionary Just 'key' is NE for primary or foreign keys Just data type is TV for the second example

3. M/J 26/P13/Q4/a,b,c

4(a)	1 mark each 3 tables with suitable names e.g. Student, Book, Borrow - All tables given have a suitable PK - Book has fields: author, title, category, number of pages - Student has fields: FName LName, class - Borrow has fields: book PK, student PK, date borrowed, field to identify book is late - FKs in borrow are clearly identified as the Book PK and the Student PK e.g. Book(<u>BID</u>, Author, Title, Category, Pages) Student(<u>SID</u>, FName, LName, Class) Borrow(<u>BwlD</u>, <u>StudentID</u>, <u>BookID</u>, DateBorrowed, Late)	6	Allow composite primary key for Borrow table. Borrow table must contain sensible linking related fields Do not accept Title or Name as PKs. Allow alternatives to late e.g. DateBorrowed, DateDueBack, DateReturned.
4(b)	1 mark each to max 2 - Ensure consistency of data // up-to-date // not redundant data - Ensure relationships between data remain valid and prevent orphaned records - If a change is made in one place it is made everywhere else // cascading updates/deletes - Every foreign key has a corresponding primary key	2	
4(c)(i)	1 mark for each error line and correction - Line 01 CREATE DATABASE SCHOOLDATA; - Line 03 SubjectCode VARCHAR NOT NULL, - Line 04 SubjectName VARCHAR, - Line 07 FOREIGN KEY (DepartmentID) REFERENCES DEPARTMENT (DepartmentID)	4	Do not award mark for Line 04 if extra Not Null has been added
4(c)(ii)	1 mark for each correct point - Selecting department name and counting subject code ... - Using suitable field name - From statement where with join / other join method - Grouping by department name / department ID - Ordering by department name in ascending order Example 1: <pre> SELECT DepartmentName, COUNT(SubjectCode) (MP1) AS "NumberSubjectCodes" (MP2) FROM DEPARTMENT, SUBJECT WHERE DEPARTMENT.DepartmentID = SUBJECT.DepartmentID (MP3) GROUP BY (DepartmentID) (MP4) ORDER BY DepartmentName; (MP5) </pre> Example 2: <pre> SELECT DepartmentName, Count(Subject.SubjectCode) (MP1) AS CountOfSubjectCode (MP2) FROM DEPARTMENT (INNER) JOIN SUBJECT ON DEPARTMENT.DepartmentID = SUBJECT.DepartmentID (MP3) GROUP BY DEPARTMENT.DepartmentName (MP4) ORDER BY (DEPARTMENT.DepartmentName) ASC; (MP5) </pre> Please note that these two lines are equivalent to MP1 and MP2 <pre> SELECT DepartmentName, NumberSubjectCodes = Count(SUBJECT.subjectcode) </pre> MP1: SELECT DepartmentName, and a correct COUNT function MP2: CountOfSubject =	5	Accept == for = Accept any correct method of joining the tables, but no mix and match between methods In example 1 the join mark needs correct FROM and WHERE clauses In example 2 the join mark needs correct FROM... JOIN and ON clauses

4. O/N 25/P11/Q2/a,b,c,d

2(a)	1 mark for each correct relationship, max 2 marks 	2
2(b)	1 mark per bullet point, max 2 marks • There are no repeating groups of attributes • There are no many-to-many relationships • There are no partial key dependencies // no non-key dependencies // no transitive dependencies	2
2(c)(i)	1 mark per bullet point, max 2 marks • DELETE FROM and correct table • Correct condition Example answer: DELETE FROM PLACEMENT WHERE Complete = TRUE;	2
2(c)(ii)	1 mark per bullet point, max 4 marks • SELECT COUNT any appropriate field • AS and an appropriate name • FROM and correct table and WHERE and one correct condition • Two AND clauses with the other two correct conditions Example answer: SELECT COUNT (CompanyID) AS TotalPlacements FROM PLACEMENT WHERE CompanyID = "NEAM" AND StudentID = "LDEA01" AND Complete = TRUE;	4

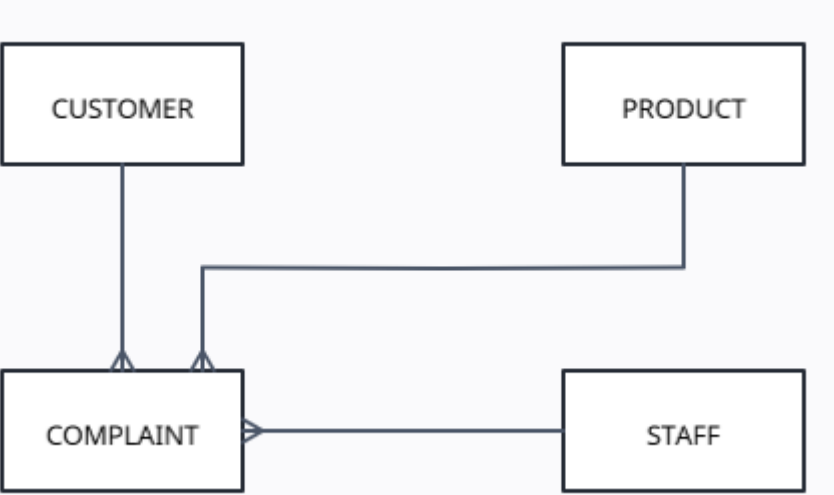
2(d)	1 mark per bullet point, max 3 marks Generic mark points (max 2) • Referential integrity ensures that related data is consistent • Referential integrity ensures that every foreign key has a corresponding primary key • Referential integrity provides for cascading update / delete • Referential integrity ensures that if a primary key is deleted or modified all linked records in foreign table will be deleted or modified // Referential integrity stops "orphaned records"—records in a table that point to an entry in another table that no longer exists Specific mark points (max 2) for good examples e.g. • CompanyID is a foreign key in PLACEMENT table and is dependent on the primary key CompanyID in COMPANY table • If a record is deleted from the STUDENT table, then all records with the same StudentID will be deleted from the PLACEMENT table • If a CompanyID is modified in the company table, all records with that CompanyID in the placement table will also be modified	3
------	---	----------

5. O/N 25/P12/Q4/a,b,c,d

4(a)	1 mark per bullet point, max 2 marks • The relationship between SHIP and CONTAINER is one-to-many (1:M) • The primary key <u>ShipID</u> in the SHIP table is linked to the foreign key ShipID in the CONTAINER table	2
4(b)	1 mark per bullet point, max 2 marks • ALTER TABLE statement • Including the additional field with a suitable field name and suitable field type Example answer <pre>ALTER TABLE CONTAINER ADD InspectionDate DATE;</pre>	2

4(c)	1 mark per bullet point, max 4 marks • SELECT <u>COUNT</u> statement • Using the correct tables • Joining the tables • The condition for the name Example Answer 1: <pre>SELECT COUNT(ContainerID) FROM CONTAINER, SHIP WHERE CONTAINER.ShipID = SHIP.ShipID AND ShipName = "Caledonia";</pre> Example Answer 2: <pre>SELECT COUNT(ContainerID) FROM CONTAINER INNER JOIN SHIP ON CONTAINER.ShipID = SHIP.ShipID WHERE ShipName = "Caledonia";</pre>	4
4(d)	1 mark per bullet point, max 2 marks • To allow the user to create / modify / delete tables // maintain the database • To allow the user to set up / modify relationships • To allow the user to create a form for data input • To allow a user to add tools to a form // for example, drop-down boxes / buttons etc. • To allow a user to design a report (to show the output in an organised manner) • To allow a user to add a menu to enable a choice of different actions • To allow a user to inspect the database contents and metadata • ... by creating and/or running different SQL queries	2

6. O/N 25/P13/Q5/a,b,c,d

5(a)	1 mark for each correct relationship, max 3 marks  <pre> graph TD CUSTOMER --> COMPLAINT PRODUCT --> COMPLAINT COMPLAINT --> STAFF </pre>	3
------	--	----------

5(b)	1 mark per bullet point, max 4 marks • Create table with opening and closing brackets and commas separating the attributes • Appropriate data types for StaffFirstName, StaffLastName and Department • Appropriate data types for StaffID and RemoteWorker • Primary key correctly defined Example Answer 1: <pre>CREATE TABLE STAFF(StaffID INTEGER, StaffFirstName VARCHAR, StaffLastName VARCHAR, Department CHAR, RemoteWorker BOOLEAN, PRIMARY KEY (StaffID));</pre> Example Answer 2: <pre>CREATE TABLE STAFF(StaffID INTEGER NOT NULL PRIMARY KEY, StaffFirstName VARCHAR, StaffLastName VARCHAR, Department CHAR, RemoteWorker BOOLEAN);</pre>	4
5(c)	1 mark per bullet point, max 5 marks • SELECT and the correct attributes • FROM and the correct tables • Tables joined correctly • Correct condition for the rating • Correct ORDER BY clause Example Answer 1 <pre>SELECT PRODUCT.ProductID, ProductName, ComplaintDetails FROM PRODUCT, COMPLAINT WHERE PRODUCT.ProductID = COMPLAINT.ProductID AND Rating <=5 ORDER BY Rating DESC;</pre> Example Answer 2 <pre>SELECT PRODUCT.ProductID, ProductName, ComplaintDetails FROM PRODUCT INNER JOIN COMPLAINT ON PRODUCT.ProductID = COMPLAINT.ProductID WHERE Rating <=5 ORDER BY Rating DESC;</pre>	5

5(d)	One mark for identification of method one mark for corresponding description max 4 marks • Authentication methods / passwords / biometrics / 2-factor authentication can be implemented • ... which prevents unauthorised access to the customer's data • Access rights / privileges can be set • ... so that only those with correct permissions can read / edit customer's data • Regular backups can be scheduled • ... so that a second copy of the customer's data is available in case of loss/damage • The data can be encrypted • ... so that the customer's data cannot be understood by anyone who gains unauthorized access • Different views can be created • ... so that not everyone can see the customer's data	4
------	---	----------

7. M/J 25/P11/Q5/a,b,c,d,e

5(a)	1 mark for each relationship. Max 2 if any extra <pre>graph TD; CUSTOMER --> CUSTOMER_CARD_DATA; CUSTOMER --> ORDER; ORDER --> ORDER_ITEM;</pre>	3								
5(b)	1 mark for: CardNumber	1								
5(c)	1 mark per row to max 2 <table><thead><tr><th>Table</th><th>Foreign key</th></tr></thead><tbody><tr><td>ORDER_ITEM</td><td>OrderID</td></tr><tr><td>ORDER</td><td>CustomerID</td></tr><tr><td>CUSTOMER_CARD_DATA</td><td>CustomerID</td></tr></tbody></table>	Table	Foreign key	ORDER_ITEM	OrderID	ORDER	CustomerID	CUSTOMER_CARD_DATA	CustomerID	2
Table	Foreign key									
ORDER_ITEM	OrderID									
ORDER	CustomerID									
CUSTOMER_CARD_DATA	CustomerID									

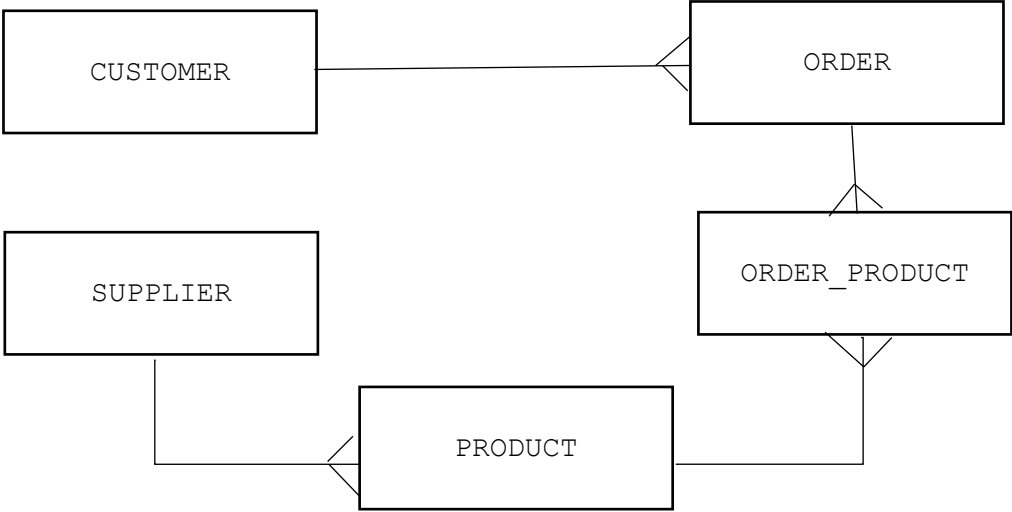
5(d)	1 mark each to max 3 e.g. - Each order would only be able to have one item - or the database would not be normalised - it would not be in 1NF - due to repeated groups of attributes - in the ORDER table	3
5(e)	1 mark each - Selecting the customer ID, customer name and sum of TotalCost with appropriate identifier - FROM clause with suitable join of tables (ON or WHERE) - ORDER.Paid = FALSE condition (with correct key word) - GROUP BY condition e.g. SELECT CUSTOMER.CustomerID, CUSTOMER.Name, Sum(ORDER.TotalCost) AS TotalOwed FROM CUSTOMER INNER JOIN ORDER ON CUSTOMER.CustomerID = ORDER.CustomerID WHERE ORDER.Paid = FALSE GROUP BY CUSTOMER.CustomerID; Alternative JOIN Statement SELECT CUSTOMER.CustomerID, CUSTOMER.Name, Sum(ORDER.TotalCost) AS TotalOwed FROM CUSTOMER,ORDER WHERE CUSTOMER.CustomerID = ORDER.CustomerID AND ORDER.Paid = FALSE GROUP BY CUSTOMER.CustomerID;	4

8. M/J 25/P12/Q5/a,b,c,d

5(a)	1 mark each to max 2 - To remove the many-to-many relationship - between CHARACTER and ITEM tables - To allow each character to have many items // to allow each item to be purchased for many characters - by creating a linking table - between the characters and the items purchased for each one	2
5(b)	1 mark for the three single Primary Keys 1 mark for the composite key USER(<u>Username</u> , Password, DateOfBirth) CHARACTER(<u>CharacterName</u> , <u>CharacterID</u> , Username, Level, Money) ITEM(<u>ItemName</u> , MinimumLevel, Cost) CHARACTER_ITEM(<u>CharacterID</u> , <u>ItemName</u>)	2

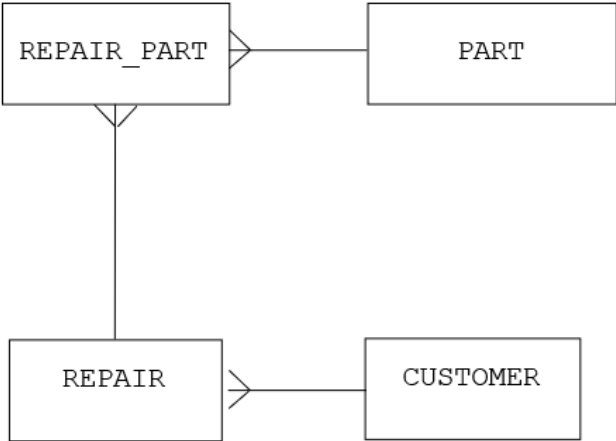
5(c)(i)	1 mark for method 1 mark for matching explanation e.g. • Access rights • appropriate permissions for the table USER are needed to read or edit the data • A password for the database or the USER table • prevents users without the password from accessing the data • Encrypting the database • stops users without the decryption key from decoding / understanding the data • Views • users can be given a view to the database that does not include the data in the table USER	4
5(c)(ii)	1 mark each to max 2 e.g. • Validation • Enforce referential integrity • Cascade update / delete • Ensuring the database is normalised	2
5(d)(i)	1 mark for each point • Select COUNT • FROM (using correct tables) and joining • WHERE (or AND) condition for correct username Example 1: SELECT COUNT(ItemName) FROM CHARACTER_ITEM INNER JOIN CHARACTER ON CHARACTER.CharacterID = CHARACTER_ITEM.CharacterID WHERE CHARACTER.Username = "KAT123"; Example 2: SELECT COUNT(ItemName) FROM CHARACTER_ITEM, CHARACTER WHERE CHARACTER.CharacterID = CHARACTER_ITEM.CharacterID AND CHARACTER.Username = "KAT123";	3
5(d)(ii)	1 mark for each point • UPDATE character • SET level and money • WHERE condition e.g. UPDATE CHARACTER SET Level = 3, Money = 10000.00 WHERE CharacterID = "0002";	3

9. M/J 25/P13/Q6/a,b,c,d

6(a)	1 mark for each correct relationship  <pre> graph LR CUSTOMER --> ORDER SUPPLIER --> PRODUCT ORDER --> ORDER_PRODUCT PRODUCT --> ORDER_PRODUCT </pre>	4
6(b)	1 mark each • INSERT INTO PRODUCT • VALUES with opening and closing brackets • Inserting ProductID, ProductName and SupplierID correctly as strings including quotation marks into correct fields • Inserting quantity and cost correctly into correct fields Example 1: INSERT INTO PRODUCT VALUES ("002323", "Blue ball point 2 mm", 50, 5.00, "SFX223"); Example 2: INSERT INTO PRODUCT (ProductID, ProductName, QuantityInBox, Cost, SupplierID) VALUES ("002323", "Blue ball point 2 mm", 50, 5.00, "SFX223");	4

6(c)	1 mark each • Selection of customer name and counting any field from ORDER as an appropriate identifier • Joining tables ORDER and CUSTOMER • AND (or WHERE) clause: Collected = FALSE • Grouping by customer ID or customer name Example 1: SELECT CustomerName, COUNT(OrderID) AS NotCollected FROM ORDER, CUSTOMER WHERE ORDER.CustomerID = CUSTOMER.CustomerID AND Collected = FALSE GROUP BY CUSTOMER.CustomerID Example 2: SELECT CustomerName, COUNT(OrderID) AS NotCollected FROM ORDER INNER JOIN CUSTOMER ON ORDER.CustomerID = CUSTOMER.CustomerID WHERE Collected = FALSE GROUP BY CUSTOMER.CustomerID	4
6(d)(i)	1 mark each to max 3 e.g. • Relationships • Views • Data types • Validation rules	3
6(d)(ii)	1 mark each to max 3 e.g. • To create / modify / delete database objects • To create a form for data input • To add tools to a form • for example, drop-down boxes / buttons etc. • To design a report to show the output in an organised manner • To add a menu to enable users to choose different actions / run different queries	3

10. O/N 24/P11/Q2/a,b,c

2(a)	1 mark for limitation and 2 marks for corresponding explanation e.g. Data redundancy // data duplication • Separate linked tables are used • Data items are stored once so reduces data duplication / data redundancy Data inconsistency / Poor data integrity • Data changed once in one place will automatically update elsewhere • Linked data cannot be entered differently in two tables • Referential integrity can be enforced The data structure used depends on the application • Changes to the data structure are managed by the DBMS • ... and queries are not dependent on the structure of the data • ... and changes to the data do not require programs to be re-written	3
2(b)(i)	1 mark for each correct relation 	3
2(b)(ii)	1 mark for each bullet point • Create table REPAIR_PART with opening and closing brackets, all statements inside the brackets • PartID and RepairNumber as varchar (or equivalent) • Quantity as integer • At least 1 appropriate constraint • Dual primary key For example: <pre>CREATE TABLE REPAIR_PART(PartID VARCHAR(20) NOT NULL, RepairNumber VARCHAR(4) NOT NULL, Quantity INT NOT NULL, PRIMARY KEY (PartID, RepairNumber));</pre>	5

2(b)(iii)	1 mark for each bullet point • Select SUM of AmountDue • From the correct table and one correct condition • Second correct condition <pre>SELECT SUM(AmountDue) FROM INVOICE WHERE SupplierID = "JK675" AND Paid = FALSE;</pre>	3								
2(c)	1 mark for each correct definition <table><tr><th>Database Term</th><th>Definition</th></tr><tr><td>Referential Integrity</td><td>All duplicate entries of data between tables are consistent // all foreign keys are matched to an appropriate primary key</td></tr><tr><td>Candidate Key</td><td>A field that could be a primary key but is not // an attribute or smallest set of attributes in a table where no tuple has the same value</td></tr><tr><td>Tuple</td><td>A row / record in a table // one instance of an entity in a table</td></tr></table>	Database Term	Definition	Referential Integrity	All duplicate entries of data between tables are consistent // all foreign keys are matched to an appropriate primary key	Candidate Key	A field that could be a primary key but is not // an attribute or smallest set of attributes in a table where no tuple has the same value	Tuple	A row / record in a table // one instance of an entity in a table	3
Database Term	Definition									
Referential Integrity	All duplicate entries of data between tables are consistent // all foreign keys are matched to an appropriate primary key									
Candidate Key	A field that could be a primary key but is not // an attribute or smallest set of attributes in a table where no tuple has the same value									
Tuple	A row / record in a table // one instance of an entity in a table									

11. O/N 24/P12/Q6/a,b

6(a)	1 mark each to max 3 e.g. - There is reduced data redundancy // less repeated data ... because each item of data is only stored once - Data consistency is maintained // Data integrity is improved ... changes in one table will automatically update in another ... linked data cannot be entered differently in two tables - Program-data independence is ensured ... changes to the data do not require programs to be re-written // queries are not dependent on the structure of the data - Complex queries are easier to run - Different views can be provided so users can only see specific aspects of the database - Multiple concurrent access is possible ... through record locking	3
------	---	----------

6(b)(i)	1 mark each to max 6 - CUSTOMER to JOB is 1 to many ... implemented by Primary Key in CUSTOMER is Foreign Key in JOB - EMPLOYEE to LOGIN_DATA is 1 to 1 ... implemented by Primary Key in EMPLOYEE is Foreign Key in LOGIN_DATA - JOB to JOB_EMPLOYEE is 1 to many ... implemented by Primary Key in JOB is Foreign Key in JOB_EMPLOYEE - EMPLOYEE to JOB_EMPLOYEE is 1 to many ... implemented by Primary Key in EMPLOYEE is Foreign Key in JOB_EMPLOYEE	6
6(b)(ii)	1 mark each - Select SUM of Amount - From the correct table and one correct condition - Remaining correct condition Example: <pre>SELECT SUM(Amount) FROM INVOICE WHERE Paid = "Y" AND DateSent >= #01/01/2023# AND DateSent <= #31/12/2023#</pre>	3

12. O/N 24/P13/Q4/a,b,c,d,e

4(a)	1 mark each: - Foreign key: BatchID, table BATCH - Foreign key: CustomerID, table CUSTOMER	2
4(b)	1 mark each: - SELECT SUM(Quantity) - FROM SALE WHERE and one correct condition - AND with remainder correct conditions e.g. <pre>SELECT SUM(Quantity) FROM SALE WHERE CustomerID = "0034E" AND Date >= #01/01/2023# AND Date <= #31/12/2023#;</pre>	3

4(c)(i)	1 mark for each bullet point • Create table BATCH with opening and closing brackets, all statements within brackets • BatchID, Type and Flavour as varchar or equivalent • ...with suitable constraint(s) • Size as decimal, SellingPrice as currency, EndDate as date (or equivalent) • Primary key identified as BatchID e.g. <pre>CREATE TABLE BATCH (BatchID VARCHAR(6) NOT NULL, Type VARCHAR (20) NOT NULL, Flavour VARCHAR (20) NOT NULL, Size FLOAT, SellingPrice CURRENCY, EndDate DATE, PRIMARY KEY (BatchID));</pre>	5
4(c)(ii)	1 mark for each bullet point each (max 4) e.g. • Ice cream table with an appropriate name • ... containing type, flavour, size, selling price • ... with suitable primary key • ... foreign key identified in BATCH that links to the primary key in ICECREAM Example table definitions – not example answer: <pre>BATCH(BatchID, IceCreamID, EndDate) ICE_CREAM(IceCreamID, Type, Flavour, Size, SellingPrice)</pre>	4
4(d)	1 mark for each definition Entity: • A real-life object that is represented as a table Attribute: • An item of data about an entity	2
4(e)	1 mark for each bullet point each (max 3) • Referential integrity is enforced • ... such as cascade update/delete // if the data is changed in one place it is updated in every other place • ... and ensures each foreign key has a corresponding primary key	3

13. M/J 24/P11/Q6/a,b,c

6(a)	1 mark each: • User table with the username as the Primary Key • ... containing at least email address, date of birth / age and rating • Quiz table with Quiz ID or date or file name as the Primary Key. • ... containing at least the other field(s) not used as the PK • A joining table with an appropriate name including at least fields for user identification, quiz identification and score • ... with an appropriate Primary Key • ... and Foreign Keys matching the Primary Keys of the other two tables USER(<u>Username</u>, Email, DateOfBirth, Rating) QUIZ(<u>QuizID</u>, Date, Filename) USER QUIZ(<u>Username</u>, <u>QuizID</u>, Score)	6
6(b)	1 mark each to max 2 for data dictionary and max 2 for logical schema: Data dictionary: • Data about the data in the database // metadata • Identifies the characteristics of the data that will be stored • Appropriate example e.g. field names, table name, validation rules, data types, primary / foreign keys, relationships etc. Logical schema: • Conceptual design • Platform/database independent overview of the database • Is used to design the physical structure • Appropriate example e.g. Design of entities / E-R diagram / views	4
6(c)(i)	1 mark for each correct clause: • Alter table EVENT • Adding foreign key as PlayerID referencing correct table ALTER TABLE EVENT ADD FOREIGN KEY(PlayerID)REFERENCES PLAYER(PlayerID);	2
6(c)(ii)	1 mark each: • Selecting PlayerID from EVENT • Counting EventID • Grouping by the PlayerID Example: SELECT PlayerID, COUNT(EventID) FROM EVENT GROUP BY PlayerID;	3

14. M/J 24/P12/Q4/a,b,c,d

4(a)	1 mark for: 1-to-many	1
4(b)	1 mark each: • Creating table EXAM with opening and closing brackets • All fields with appropriate data types and commas at end of lines • ExamID as primary key Example: <pre>CREATE TABLE EXAM(ExamID varchar NOT NULL, Subject varchar, Level int, TotalMarks int, PRIMARY KEY (ExamID));</pre>	3
4(c)	1 mark each: • Altering table EXAM_QUESTION • Linking ExamID to ExamID in EXAM Example. <pre>ALTER TABLE EXAM_QUESTION ADD FOREIGN KEY (ExamID) REFERENCES EXAM (ExamID);</pre>	2
4(d)	1 mark each to max 5: • STUDENT table identified with suitable Primary Key • A linking table between STUDENT and EXAM with suitable Primary Key and appropriate name • ... that includes the Primary Key of the STUDENT table as a Foreign Key to join with STUDENT • ... and includes the Primary Key of the EXAM table as a Foreign Key to join with EXAM • A linking table between STUDENT and EXAM_QUESTION with suitable Primary Key and appropriate name • ... that includes the Primary Key of Table 2 as a Foreign Key to join with Table 2 • ... that stores the ExamQuestionID and the mark for that question	5

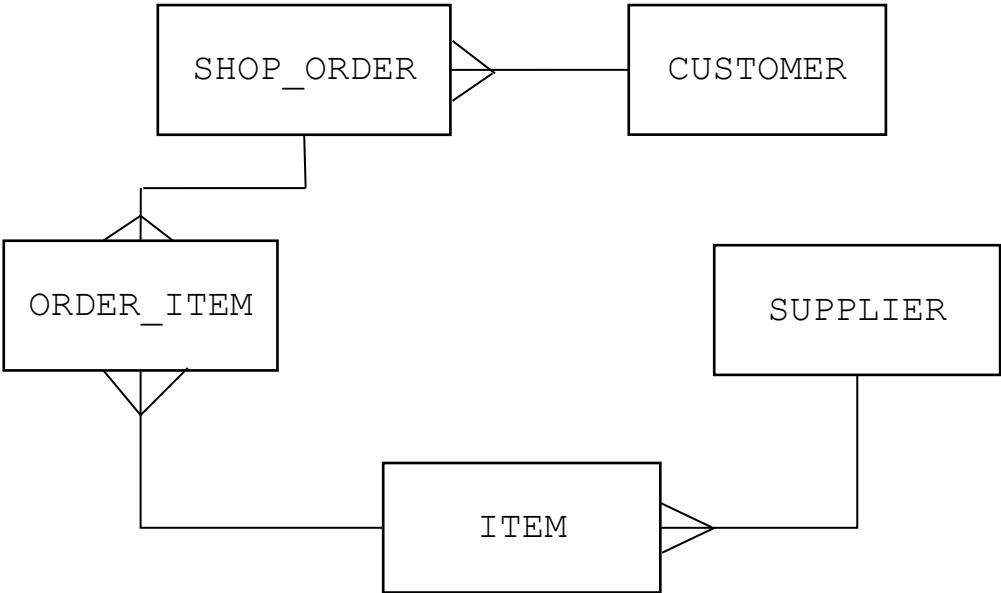
15. M/J 24/P13/Q4/a,b,c,d

4(a)	1 mark for: many to 1 // there are many performances of each show	1
------	---	----------

4(b)	1 mark each for: • Creating table <code>PERFORMANCE</code> with opening and closing brackets • Setting all four attributes with appropriate data types • Setting <code>PerformanceID</code> as primary key • Setting <code>ShowID</code> as foreign key referencing <code>SHOW</code> table Example: <pre>CREATE TABLE PERFORMANCE(PerformanceID varchar NOT NULL, ShowID varchar, ShowDate Date, StartTime Time, PRIMARY KEY (PerformanceID), FOREIGN KEY (ShowID) REFERENCES SHOW (ShowID));</pre>	4
4(c)	1 mark each for: • Selecting <code>COUNT</code> of an attribute in <code>PERFORMANCE</code> table with suitable name • <code>FROM</code> clause • Joining tables • Grouping by the title and selecting the title Example 1: <pre>SELECT SHOW.Title, Count (PERFORMANCE.PerformanceID) AS NumberOfShowings FROM PERFORMANCE, SHOW WHERE PERFORMANCE.ShowID = SHOW.ShowID GROUP BY SHOW.Title;</pre> Example 2: <pre>SELECT SHOW.Title, Count (PERFORMANCE.PerformanceID) AS NumberOfShowings FROM PERFORMANCE INNER JOIN SHOW ON PERFORMANCE.ShowID = SHOW.ShowID GROUP BY SHOW.Title;</pre>	4

4(d)	1 mark each to max 5: e.g. • CUSTOMER table identified with suitable Primary Key and appropriate name • ... and other suitable fields including name and email • BOOKING TABLE identified with suitable Primary Key and appropriate name • ... that stores the Primary Key of the CUSTOMER table as a Foreign Key to join with CUSTOMER table • ... and stores the Primary Key of the PERFORMANCE table as a Foreign Key to join with PERFORMANCE table • A linking table between Table 2 and SEAT with suitable Primary Key and appropriate name • ... that includes the Primary Key of Table 2 as a Foreign Key to join with Table 2 • ... that stores the SeatID.	5
------	---	----------

16. O/N 23/P11/Q3/a,b,c

3(a)	1 mark for each correct relationship or relationships • 1:M between CUSTOMER and SHOP_ORDER • 1:M between SUPPLIER and ITEM • 1:M between SHOP_ORDER and ORDER_ITEM and M:1 between ORDER_ITEM and ITEM  <pre> graph TD SHOP_ORDER[SHOP_ORDER] --> 1:M CUSTOMER[CUSTOMER] SHOP_ORDER --> 1:M ORDER_ITEM[ORDER_ITEM] ORDER_ITEM --> M:1 ITEM[ITEM] SUPPLIER[SUPPLIER] --> 1:M ITEM </pre>	3
------	---	----------

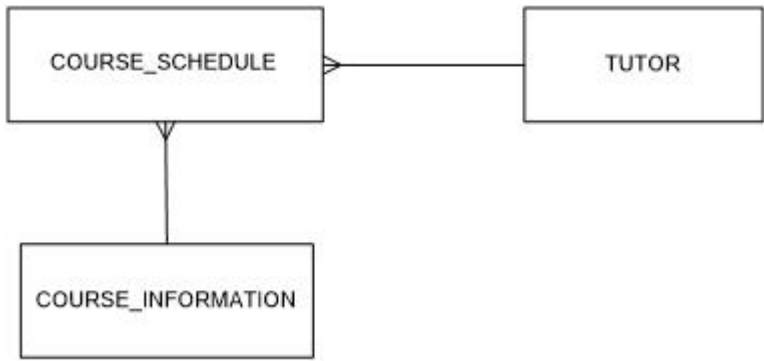
3(b)	1 mark for each bullet point (max 3) - Reduced data redundancy - Improved data integrity / consistency / referential integrity - Allows for views / improved privacy - Allows for program-data independence - Complex queries can be executed	3
3(c)(i)	1 mark for CREATE DATABASE SHOP;	1
3(c)(ii)	1 mark for each line (max 4) Either: SELECT SUM(Quantity) FROM ORDER_ITEM, SHOP_ORDER WHERE ORDER_ITEM.OrderNo = SHOP_ORDER.OrderNo AND SHOP_ORDER.CustomerID = 'HJ231'; OR SELECT SUM(Quantity) FROM ORDER_ITEM (INNER) JOIN SHOP_ORDER ON ORDER_ITEM.OrderNo = SHOP_ORDER.OrderNo WHERE SHOP_ORDER.CustomerID = 'HJ231';	4

17. O/N 23/P12/Q2/a,b,c

2(a)	1 mark for each term (max 3) Entity: - An object about which data can be stored Primary key: - The unique attribute / combination of attributes used to identify the record / tuple Referential integrity: - Makes sure that if data is changed in one place the change is reflected in all related records - cascading update/delete - Makes sure that data that does not exist cannot be referenced - Ensures that every foreign key has a corresponding primary key // A logical dependency of a foreign key on a primary key - Ensures that the data in the database is consistent / up to date - Prevents records from being added/deleted/modified incorrectly - Makes sure any queries return accurate and complete results	3
------	--	----------

2(b)	1 mark for each bullet point (max 4) Max 2 if no descriptions • Backup / recovery procedures • ... automatically takes copies of the database and store off site on a regular basis / weekly, etc. • ... so that the data can be recovered if lost • Use of access rights • ... some users are given different access permissions to different tables • ... read/write, read only, full access, etc. • Views • ... different users are able to see different parts of the database • ... only see what users need to see // by example • Record and table locking • ... prevents simultaneous access to data • ... so updates are not lost // data is not overwritten • Encryption • ... the data is turned into ciphertext • ... so it cannot be understood without a decryption key	4
2(c)	1 mark for each bullet point (max 4) • Identify repeating groups of attributes ... • ... Subject and SubjectCode • Ensure each field is atomic • ... StudentName should be split into e.g. FirstName and LastName • Identify the primary key for the table	4

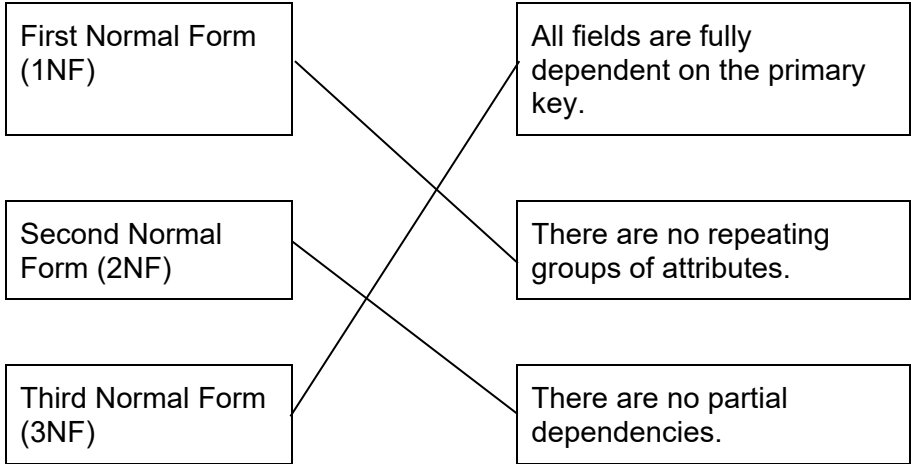
18. O/N 23/P13/Q3/b

3(b)(i)	1 mark for both 1:M relationships as follows:  <pre> graph TD CS[COURSE_SCHEDULE] --> T[TUTOR] CS --> CI[COURSE_INFORMATION] </pre>	1
---------	---	----------

3(b)(ii)	1 mark for each bullet point. • <code>SELECT Count (CourseID)</code> • <code>AS NumOfCourses</code> • <code>FROM COURSE_SCHEDULE</code> • <code>WHERE DateStarted > "09/09/23";</code> <code>SELECT Count (CourseID) AS NumOfCourses FROM COURSE_SCHEDULE WHERE DateStarted > "09/09/23";</code>	4
----------	--	----------

19. M/J 23/P11/Q2/a,b

2(a)(i)	1 mark for definition: • Data about the data in the database // data about the structure of the database // metadata for a database 1 mark for a suitable example Examples: • table names • data types • field names	2						
2(a)(ii)	1 mark for definition • Methods of making sure the data is consistent 1 mark for example Examples: • Enforcing referential integrity • If data in one table is deleted/edited all tables are updated // cascading update/delete • Validation/verification rules	2						
2(b)(i)	1 mark for each field name and table <table><tr><th>Foreign key</th><th>Database table</th></tr><tr><td>BirdID</td><td>BIRD_SEEN</td></tr><tr><td>PersonID</td><td>BIRD_SEEN</td></tr></table>	Foreign key	Database table	BirdID	BIRD_SEEN	PersonID	BIRD_SEEN	2
Foreign key	Database table							
BirdID	BIRD_SEEN							
PersonID	BIRD_SEEN							

2(b)(ii)	1 mark for all 3 correct lines Normal Form First Normal Form (1NF) Second Normal Form (2NF) Third Normal Form (3NF) Definition All fields are fully dependent on the primary key. There are no repeating groups of attributes. There are no partial dependencies. 	1
2(b)(iii)	1 mark each • CREATE TABLE start and end bracket • Bird ID as CHAR/VARCHAR • Name and size as VARCHAR/CHAR • Bird ID as primary key Example answer: <pre>CREATE TABLE BIRD_TYPE (BirdID CHAR(4) NOT NULL, Name VARCHAR(9), Size VARCHAR(6), PRIMARY KEY (BirdID));</pre>	4
2(b)(iv)	1 mark for each correctly completed space <pre>SELECT BIRD_TYPE.Size, COUNT(BIRD_TYPE.BirdID) AS NumberOfBirds FROM BIRD_TYPE, BIRD_SEEN WHERE BIRD_SEEN.PersonID = "J_123" AND BIRD_TYPE.BirdID = BIRD_SEEN.BirdID GROUP BY BIRD_TYPE.Size;</pre>	5

20. M/J 23/P12/Q2/a,b,c(ii,iii)

2(a)	1 mark for each correct feature or description <table><tr><th>Feature</th><th>Description</th></tr><tr><td>Data dictionary</td><td>Data about the data in the database // data about the structure of the database // metadata for a database</td></tr><tr><td>Query processor</td><td>Software that allows the user to enter criteria, then finds and returns the appropriate result // software that processes and executes queries written in SQL</td></tr><tr><td><u>Logical schema</u></td><td>A model of a database that is not specific to one DBMS</td></tr><tr><td><u>Developer interface</u></td><td>A software tool that allows the user to create items such as tables, forms and reports</td></tr></table>	Feature	Description	Data dictionary	Data about the data in the database // data about the structure of the database // metadata for a database	Query processor	Software that allows the user to enter criteria , then finds and returns the appropriate result // software that processes and executes queries written in SQL	<u>Logical schema</u>	A model of a database that is not specific to one DBMS	<u>Developer interface</u>	A software tool that allows the user to create items such as tables, forms and reports	4
Feature	Description											
Data dictionary	Data about the data in the database // data about the structure of the database // metadata for a database											
Query processor	Software that allows the user to enter criteria , then finds and returns the appropriate result // software that processes and executes queries written in SQL											
<u>Logical schema</u>	A model of a database that is not specific to one DBMS											
<u>Developer interface</u>	A software tool that allows the user to create items such as tables, forms and reports											
2(b)	1 mark each to max 3 • Referential Integrity makes sure data is consistent • Referential Integrity makes sure all data is up-to-date • Referential integrity ensures that every foreign key has a corresponding primary key • Referential Integrity prevents records from being added / deleted / modified incorrectly • Referential Integrity makes sure that if data is changed in one place the change is reflected in all related records • Referential Integrity makes sure any queries return accurate and complete results	3										
2(c)(ii)	1 mark each • SELECT field Name • FROM table HORSE • WHERE with Intermediate / Beginner • OR with Beginner / Intermediate Example answer: SELECT Name FROM HORSE WHERE HorseLevel = "Intermediate" OR HorseLevel = "Beginner";	4										

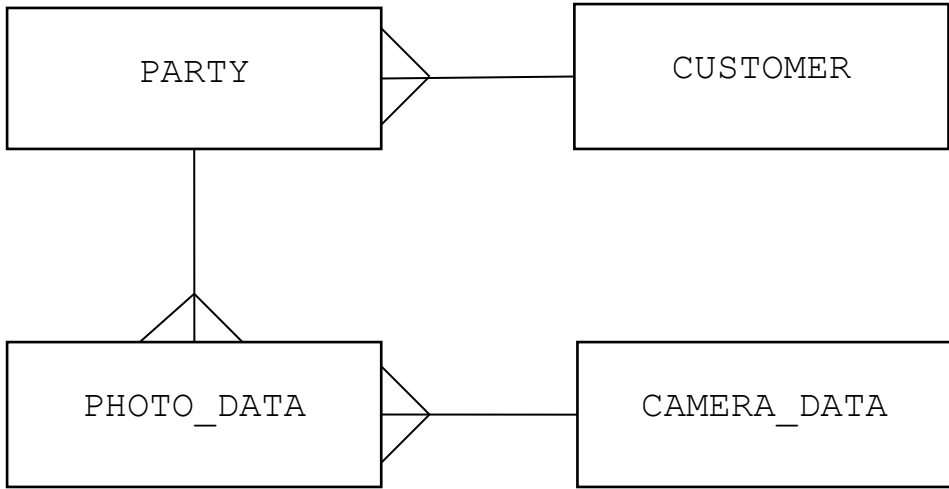
2(c)(iii)	1 mark each - SUM should be COUNT // SELECT COUNT(STUDENT.RiderLevel) - The WHERE statement needs the table names before each field name // WHERE STUDENT.StudentID = LESSON.StudentID - The OR should be AND // AND Date = #09/09/2023# - Beginner is missing the speech marks // STUDENT.RiderLevel = "Beginner";	4
-----------	---	----------

21. M/J 23/P13/Q4/a,b,c

4(a)	1 mark for each bullet point. Mark in pairs. Max 2 for each description. • Reduces data redundancy • ... because linked tables mean that each data item is stored only once • Reduces program-data dependency • ... because the data is separate from the software so changes to the data do not require programs to be re-written • Reduces data inconsistency // improves data integrity • ... because by only storing data once it only needs to be updated once // changes in one table will automatically update in another // linked data cannot be entered differently in two tables • Complex queries are easier to run • Can provide different views •so users can only see specific aspects of the database	4										
4(b)	1 mark each <table><tr><th>Term</th><th>Description</th></tr><tr><td>Entity</td><td>An object that data is stored about.</td></tr><tr><td>Tuple</td><td>A row of data in a table about one instance of an object.</td></tr><tr><td>Secondary key</td><td>An additional/alternative key used as well as the primary key to locate specific data // a candidate key that has not been chosen as a primary key.</td></tr><tr><td>Foreign key</td><td>A field in one table that is linked to a primary key in another table.</td></tr></table>	Term	Description	Entity	An object that data is stored about.	Tuple	A row of data in a table about one instance of an object.	Secondary key	An additional/alternative key used as well as the primary key to locate specific data // a candidate key that has not been chosen as a primary key.	Foreign key	A field in one table that is linked to a primary key in another table.	4
Term	Description											
Entity	An object that data is stored about.											
Tuple	A row of data in a table about one instance of an object.											
Secondary key	An additional/alternative key used as well as the primary key to locate specific data // a candidate key that has not been chosen as a primary key.											
Foreign key	A field in one table that is linked to a primary key in another table.											

4(c)	1 mark each - Only 3 tables with appropriate identifiers (i.e. one table for customer, one for booking and one for car) - Appropriate Primary key in each table underlined - Booking table includes Primary key from car and Primary key from customer as Foreign keys - All original fields are in correct tables Example answer: BOOKING(<u>BookingID</u>, CarRegistration, CustomerID, StartDate, EndDate) CAR(<u>CarRegistration</u>, CarModel, CarColour) CUSTOMER(<u>CustomerID</u>, CustomerFirstName, CustomerLastName, EmailAddress, TelephoneNumber)	4
------	--	----------

22. O/N 22/P11/Q4/a,b,c,d

4(a)	1 mark for each correct relationship: 1:M between CUSTOMER and PARTY 1:M between PARTY and PHOTO_DATA 1:M between CAMERA_DATA and PHOTO_DATA 	3
4(b)	1 mark for each bullet point (max 3): - no repeating groups of attributes // data is atomic - no partial key dependencies - no non-key dependencies // no transitive dependencies	3
4(c)(i)	1 mark for the definition, 1 mark for the example: - definition: a single row in a table - example: from the PHOTO_DATA table	2

4(c)(ii)	1 mark for each correctly completed empty space: • COUNT • PhotoID • PHOTO_DATA • 'CAN*' // 'CAN%' <pre>SELECT COUNT(PhotoID) FROM PHOTO_DATA WHERE CameraID LIKE 'CAN*'; // WHERE CameraID LIKE 'CAN%';</pre>	4
4(d)	1 mark for each bullet point: • ALTER TABLE CAMERA_DATA • ADD NumberStored INTEGER • , LastUsed DATE; <pre>ALTER TABLE CAMERA_DATA ADD NumberStored INTEGER, LastUsed DATE;</pre>	3

23. O/N 22/P12/Q5/a,b,c,d

5(a)	1 mark for each bullet point (max 3): Solution 1: • removing the many-to-many relationship between Owner and Tree • ... by removing TreeID and TreePosition from the Owner table • ... and creating a linking table between Owner and Tree • ... that contains OwnerID, TreeID and TreePosition • ... (composite) primary key of the linking table should be OwnerID and TreeID // insert a named new primary key in the linking table Solution 2: • removing the many-to-many relationship between Owner and Tree • move TreePosition into TREE table • ... put OwnerID into TREE table • create a new table with suitable name (for the species of tree) • ... containing ScientificName, MaxHeight and FastGrowing • ... with ScientificName as primary key // or another suitable primary key	3
------	---	----------

5(b)	1 mark for each bullet point: • INSERT INTO TREE • VALUES () and correct values • Values in correct order Option 1: INSERT INTO TREE(TreeID, ScientificName, MaxHeight, FastGrowing) VALUES('LOW_1276', 'Salix_Alba', 30.00, TRUE); Option 2: INSERT INTO TREE VALUES('LOW_1276', 'Salix_Alba', 30.00, TRUE);	3
5(c)	1 mark for: An attribute / field (or set of attributes / fields) that could be a primary key	1
5(d)(i)	1 mark for description • stores metadata about the database 1 mark for a correct example For example: • field / attribute names • table name • validation rules • data types • primary keys // foreign keys • relationships	2
5(d)(ii)	1 mark for each bullet point (max 2): • the overview of a database structure • models the problem / situation • ... by using methods such as an ER diagram • independent of any particular DBMS	2

24. O/N 22/P13/Q2/a,b,c,d,f

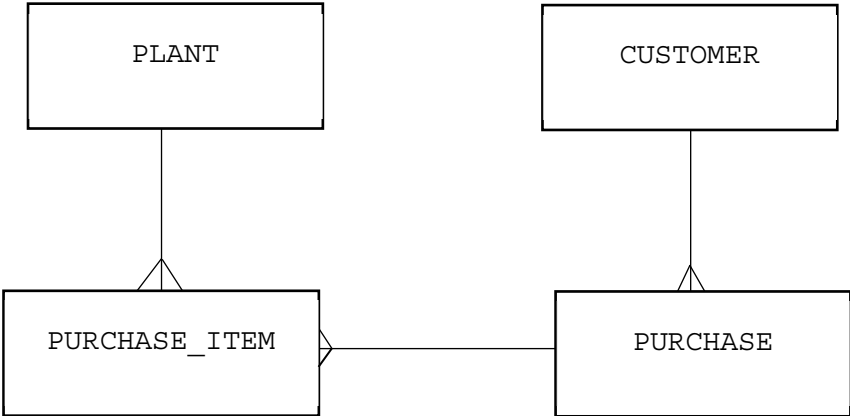
2(a)	1 mark for each correct answer. <table><thead><tr><th>Item</th><th>Answer</th></tr></thead><tbody><tr><td>a suitable field for the primary key in COMPANY</td><td>CompanyID</td></tr><tr><td>a candidate key in TELESCOPE</td><td>SerialNumber // TelescopeID</td></tr><tr><td>the degree of relationship between TELESCOPE and PHOTOGRAPH</td><td>1:M / 1 to many</td></tr></tbody></table>	Item	Answer	a suitable field for the primary key in COMPANY	CompanyID	a candidate key in TELESCOPE	SerialNumber // TelescopeID	the degree of relationship between TELESCOPE and PHOTOGRAPH	1:M / 1 to many	3
Item	Answer									
a suitable field for the primary key in COMPANY	CompanyID									
a candidate key in TELESCOPE	SerialNumber // TelescopeID									
the degree of relationship between TELESCOPE and PHOTOGRAPH	1:M / 1 to many									
2(b)	Logical schema	1								

2(c)	1 mark for each correctly completed missing part: SELECT COUNT (TelescopeID) FROM TELESCOPE WHERE CompanyID LIKE 'HW%' ;	4
2(d)	1 mark for each bullet point: - ALTER TABLE PHOTOGRAPH - ADD Resolution TEXT; ALTER TABLE PHOTOGRAPH ADD Resolution TEXT / VARCHAR(11);	2
2(f)	1 mark for each bullet point (max 2): - allows the user to enter criteria - searches for the data that meets the entered criteria - organises the results to be displayed to the user	2

25. O/N 21/P11/Q5/a,b,c

5(a)	1 mark for 2 or 3 correct ticks, 2 marks for 4 correct ticks	2																				
	<table><tr><th>Table</th><th>Field name</th><th>Primary Key (PK)</th><th>Foreign Key (FK)</th></tr><tr><td>MANAGER</td><td>ManagerID</td><td>✓</td><td></td></tr><tr><td>SHOP</td><td>ManagerID</td><td></td><td>✓</td></tr><tr><td>CAR</td><td>RegistrationNumber</td><td>✓</td><td></td></tr><tr><td>CAR</td><td>ShopID</td><td></td><td>✓</td></tr></table>	Table	Field name	Primary Key (PK)	Foreign Key (FK)	MANAGER	ManagerID	✓		SHOP	ManagerID		✓	CAR	RegistrationNumber	✓		CAR	ShopID		✓	
Table	Field name	Primary Key (PK)	Foreign Key (FK)																			
MANAGER	ManagerID	✓																				
SHOP	ManagerID		✓																			
CAR	RegistrationNumber	✓																				
CAR	ShopID		✓																			
5(b)	1 mark per bullet point - Access rights give managers / himself access to different elements ... by having different accounts / logins ... which have different access rights e.g. read only // no access / read / write - Specific <u>views</u> can be assigned to himself and to the managers ... e.g. managers can only see the data for their own shop(s)	3																				
5(c)(i)	1 mark per correctly completed statement SELECT COUNT(RegistrationNumber) FROM CAR GROUP BY ShopID	3																				
5(c)(ii)	1 mark for each correct statement INSERT INTO CAR VALUES ("123AA", "Tiger", "Lioness", 10500, "12BSTREET")	2																				

26. O/N 21/P12/Q6/a,b,c

6(a)(i)	1 mark for 1 tick in the correct place 2 marks for all 3 ticks correct <table><thead><tr><th rowspan="2">Task</th><th colspan="3">Normalisation stage</th></tr><tr><th>0NF to 1NF</th><th>1NF to 2NF</th><th>2NF to 3NF</th></tr></thead><tbody><tr><td>Remove any partial key dependencies</td><td></td><td>✓</td><td></td></tr><tr><td>Remove any repeating groups of attributes</td><td>✓</td><td></td><td></td></tr><tr><td>Remove any non-key dependencies</td><td></td><td></td><td>✓</td></tr></tbody></table>	Task	Normalisation stage			0NF to 1NF	1NF to 2NF	2NF to 3NF	Remove any partial key dependencies		✓		Remove any repeating groups of attributes	✓			Remove any non-key dependencies			✓	2
Task	Normalisation stage																				
	0NF to 1NF	1NF to 2NF	2NF to 3NF																		
Remove any partial key dependencies		✓																			
Remove any repeating groups of attributes	✓																				
Remove any non-key dependencies			✓																		
6(a)(ii)	1 mark for each correct relationship 	3																			
6(b)	1 mark for description of purpose - Stores metadata about the database 1 mark for each example of contents to max 2 e.g. - field / attribute names - table name - validation rules - data types - primary keys // foreign keys - relationships	3																			
6(c)(i)	1 mark for each correctly completed space <pre>SELECT SUM(Quantity) FROM PURCHASE_ITEM WHERE PurchaseID = "3011A";</pre>	4																			

6(c)(ii)	1 mark per bullet point - ALTER TABLE PURCHASE - ADD OrderDate - Suitable data type, e.g. DATE <pre>ALTER TABLE PURCHASE ADD OrderDate DATE;</pre>	3
----------	---	----------

27. O/N 21/P13/Q5/a,b,c

5(a)	1 mark for 2 or 3 correct ticks, 2 marks for 4 correct ticks	2																				
	<table><tr><th>Table</th><th>Field name</th><th>Primary Key (PK)</th><th>Foreign Key (FK)</th></tr><tr><td>MANAGER</td><td>ManagerID</td><td>✓</td><td></td></tr><tr><td>SHOP</td><td>ManagerID</td><td></td><td>✓</td></tr><tr><td>CAR</td><td>RegistrationNumber</td><td>✓</td><td></td></tr><tr><td>CAR</td><td>ShopID</td><td></td><td>✓</td></tr></table>	Table	Field name	Primary Key (PK)	Foreign Key (FK)	MANAGER	ManagerID	✓		SHOP	ManagerID		✓	CAR	RegistrationNumber	✓		CAR	ShopID		✓	
Table	Field name	Primary Key (PK)	Foreign Key (FK)																			
MANAGER	ManagerID	✓																				
SHOP	ManagerID		✓																			
CAR	RegistrationNumber	✓																				
CAR	ShopID		✓																			
5(b)	1 mark per bullet point - Access rights give managers / himself access to different elements ... by having different accounts / logins ... which have different access rights e.g. read only // no access / read / write - Specific <u>views</u> can be assigned to himself and to the managers ... e.g. managers can only see the data for their own shop(s)	3																				
5(c)(i)	1 mark per correctly completed statement SELECT COUNT(RegistrationNumber) FROM CAR GROUP BY ShopID	3																				
5(c)(ii)	1 mark for each correct statement INSERT INTO CAR VALUES ("123AA", "Tiger", "Lioness", 10500, "12BSTREET")	2																				

28. M/J 21/P11/Q7/a,b,c

7(a)	1 mark per bullet point to max 3 • Flat-file has more data redundancy • ... because the same data is stored many times // data is stored in different tables which are linked • There is program-data dependence with flat-files • ... because any changes to the structure of the data means the programs that access that data have to be re-written • Flat-file has more data inconsistency // worse data integrity • ... because duplicated data might be stored differently //...because when data is updated in one place, it is not updated everywhere • It is not easy to perform complex searches /queries • ... because a new program has to be written each time • Flat files could have a lack of privacy • ... as user views cannot easily be implemented	3								
7(b)(i)	1 mark for each correct example one-to-one • e.g. customer to payment details // customer to login details one-to-many • e.g. customer to order many-to-many • e.g. order to product // customer to product	3								
7(b)(ii)	1 mark <table border="1"><thead><tr><th>Relationship</th><th>Tick (✓)</th></tr></thead><tbody><tr><td>one-to-one</td><td></td></tr><tr><td>one-to-many</td><td></td></tr><tr><td>many-to-many</td><td>✓</td></tr></tbody></table>	Relationship	Tick (✓)	one-to-one		one-to-many		many-to-many	✓	1
Relationship	Tick (✓)									
one-to-one										
one-to-many										
many-to-many	✓									
7(b)(iii)	1 mark CREATE DATABASE SHOPORDERS ;	1								
7(c)	1 mark per item to max 3 • table name • field name // attribute • data type • type of validation • Primary Key • Foreign Key • relationships	3								

29. M/J 21/P12/Q1/a,b,c

1(a)	1 mark for definition, 1 mark for appropriate example in each <table><tr><td>Term</td><td>Definition and example</td></tr><tr><td>Field</td><td>A column/attribute in a table e.g. CustomerID in the table CUSTOMER</td></tr><tr><td>Entity</td><td>Anything that data can be stored about e.g. A customer or a house</td></tr><tr><td>Foreign Key</td><td>A field in one table that is linked to a Primary Key in another table e.g. CustomerID / HouseID <u>in table RENTAL</u></td></tr></table>	Term	Definition and example	Field	A column/attribute in a table e.g. CustomerID in the table CUSTOMER	Entity	Anything that data can be stored about e.g. A customer or a house	Foreign Key	A field in one table that is linked to a Primary Key in another table e.g. CustomerID / HouseID <u>in table RENTAL</u>	6
Term	Definition and example									
Field	A column/attribute in a table e.g. CustomerID in the table CUSTOMER									
Entity	Anything that data can be stored about e.g. A customer or a house									
Foreign Key	A field in one table that is linked to a Primary Key in another table e.g. CustomerID / HouseID <u>in table RENTAL</u>									
1(b)	1 mark per bullet point to max 2 - All fields in all tables are dependant fully on the PK and on no other fields - for example all fields in Customer table are fully dependent on CustomerID	2								
1(c)(i)	1 mark for each correctly completed line <pre>CREATE TABLE RENTAL (RentalID INTEGER NOT NULL, CustomerID INTEGER NOT NULL, HouseID VARCHAR (5) NOT NULL, MonthlyCost REAL/CURRENCY NOT NULL, DepositPaid BOOLEAN NOT NULL, PRIMARY KEY (RentalID));</pre>	4								
1(c)(ii)	1 mark per bullet point - Select FirstName and LastName - From both tables - Where DepositPaid = No - Joining tables (either AND, or INNER JOIN) Example script: SELECT FirstName, LastName FROM CUSTOMER, RENTAL WHERE DepositPaid = No AND RENTAL.CustomerID = CUSTOMER.CustomerID;	4								

30. M/J 21/P13/Q7/a,b,c

7(a)	1 mark per bullet point to max 3 • Flat-file has more data redundancy • ... because the same data is stored many times // data is stored in different tables which are linked • There is program-data dependence with flat-files • ... because any changes to the structure of the data means the programs that access that data have to be re-written • Flat-file has more data inconsistency // worse data integrity • ... because duplicated data might be stored differently //...because when data is updated in one place, it is not updated everywhere • It is not easy to perform complex searches /queries • ... because a new program has to be written each time • Flat files could have a lack of privacy • ... as user views cannot easily be implemented	3								
7(b)(i)	1 mark for each correct example one-to-one • e.g. customer to payment details // customer to login details one-to-many • e.g. customer to order many-to-many • e.g. order to product // customer to product	3								
7(b)(ii)	1 mark <table border="1"><thead><tr><th>Relationship</th><th>Tick (✓)</th></tr></thead><tbody><tr><td>one-to-one</td><td></td></tr><tr><td>one-to-many</td><td></td></tr><tr><td>many-to-many</td><td>✓</td></tr></tbody></table>	Relationship	Tick (✓)	one-to-one		one-to-many		many-to-many	✓	1
Relationship	Tick (✓)									
one-to-one										
one-to-many										
many-to-many	✓									
7(b)(iii)	1 mark CREATE DATABASE SHOPORDERS ;	1								
7(c)	1 mark per item to max 3 • table name • field name // attribute • data type • type of validation • Primary Key • Foreign Key • relationships	3								